UNIVERSIDAD AUTÓNOMA DE NUEVO LEÓN

FACULTAD DE INGENIERÍA MECÁNICA Y ELÉCTRICA

SUBDIRECCIÓN DE ESTUDIOS DE POSGRADO

YIELD-AWARE HIGHER-ORDER TRAJECTORY OPTIMIZATION FOR STEP-AND-SCAN WAFER STAGES

POR
ROBERTO TREVIÑO CERVANTES

COMO REQUISITO PARA OBTENER EL GRADO DE
MAESTRÍA EN CIENCIAS DE LA INGENIERÍA ELÉCTRICA

OCTUBRE 2026

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Los miembros del Comite de Tesis recomendamos que la tesis "**Yield-Aware Higher-Order Trajectory Optimization for Step-and-Scan Wafer Stages**" realizada por el(la) alumno(a) **Roberto Treviño Cervantes**, con numero de matricula 1915003, sea aceptada para su defensa como opcion al grado de **Maestría en Ciencias de la Ingenieria Electrica.**

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A ... por su apoyo ... ETC

Agradecimientos

Agradezco a ... blah blah blah.

Abstract

Step-and-scan photolithography scanners expose the layers of an integrated circuit by sweeping a reticle and a silicon wafer through a projection slit in opposite directions, synchronized to nanometer-level accuracy. The reference trajectories that command these motions determine how much energy is injected into the poorly damped structural modes of the machine, and therefore how large the residual synchronization error is during exposure. Trajectory design is conventionally evaluated through positioning metrics none of which describes the morphology of the defects that those errors ultimately imprint on the wafer.

This work proposes a yield-aware framework for optimizing the kinematic bounds of fourth-order (snap-limited) reference trajectories. A planar multi-body model of a generic lithography machine is implemented in the MuJoCo physics engine and validated against the characterization study it derives from. The synchronization error produced during scanning is transformed into a simulated wafer defect map, which a convolutional neural network (CNN) trained on the WM-811K dataset classifies into nine defect morphologies. The probability distribution estimated by the CNN, restricted to the classes plausibly caused by scanner motion, defines a cost function that links trajectory parameters directly to expected yield, closing the loop for data-driven trajectory optimization under a throughput constraint.

Keywords: photolithography, step-and-scan, trajectory generation, fourth-order profiles, multi-body dynamics, convolutional neural networks, wafer maps, yield.

Symbols and notation

$\dot{x}, \ddot{x}$	Dots denote derivatives with respect to time, e.g. $\dot{x} = dx/dt$.
x^T, A^T	Transpose of the vector x and the matrix A.
$\mathbb{R}$	The field of real numbers.
$\|x\|$	A norm of x.
$\mathcal{F}_0, \mathcal{F}_1$	World (inertial) frame and base frame of the machine.
j	Chain index: $j = 1$ reticle, $j = 2$ optics, $j = 3$ wafer.
i_j	Body i within chain j; N_j is the last body of chain j.
f_{i_j}	In-plane displacement of body i_j relative to its parent body.
$f^0_{i_j}$	Absolute in-plane displacement of body i_j relative to the inertial frame $\mathcal{F}_0$.
$\theta_{z_{i_j}}$	Rotation of body i_j about the vertical axis.
${}^0R_{i_j}$	Rotation matrix from the local frame of body i_j to the inertial frame $\mathcal{F}_0$.
$m_{i_j}, I_{z_{i_j}}$	Mass and z-axis moment of inertia of body i_j.
K_{i_j}, C_{i_j}	Stiffness and damping of the Kelvin–Voigt coupling of body i_j.
$\mathbf{x}, \mathbf{u}$	State and input vectors in the state-space representation.
u_{i_j}, τ_{i_j}	Control forces and torques applied to body i_j.
α	Projection demagnification factor, $\alpha = 0.25$.
e_{syn}	Synchronization error between wafer and reticle stages.
MA, MSD	Moving average and moving standard deviation of e_{syn} over the exposure window.
$v_{\max}, a_{\max}$	Velocity and acceleration bounds of the reference trajectory.
$j_{\max}, s_{\max}$	Jerk and snap bounds of the reference trajectory.
t_s	Settling time.
CD	Critical dimension: minimum printable feature size.
NA	Numerical aperture of the projection lens.
λ	Illumination wavelength.
P_c	Probability assigned by the CNN to defect class c.
w_c	Economic weight of defect class c in the cost function.
$\mathcal{J}$	Trajectory cost: expected yield loss, $\mathcal{J} = \sum_c w_c P_c$.
WPH	Throughput in wafers per hour.
Italics	Italics denote defect class names and emphasis on key concepts.
Boldface	Boldface denotes emphasis, structural elements, and technical terms at first use.

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Chapter 1

Introduction

The history of modern electronics is inseparable from the history of lithography. Since the independent invention of the integrated circuit (IC) by Jack Kilby at Texas Instruments and Robert Noyce at Fairchild Semiconductor in 1958, Moore's Law has governed the relentless progression of the semiconductor industry: the density of components on a chip doubles approximately every two years[13]. This exponential trend has translated into dramatic reductions in cost alongside equally dramatic increases in performance. The price of one gigabyte of NAND flash memory, for instance, fell from over \$1000 at the start of this millennium to below \$0.30 by the mid-2010s[3], while the minimum transistor feature size has shrunk from hundreds of nanometers to just a few tens of nanometers. The engine of this revolution is *photolithography* — the process by which the intricate patterns of a transistor, a wire, or a memory cell are projected onto a silicon wafer with nanometer-scale fidelity.

The fabrication of a modern IC consists of up to 30 sequential process cycles for mature technology nodes[13], each defining one distinct layer of the device; at leading-edge nodes — such as 5 nm CMOS production technology — the layer count rises to approximately 75[17;12]. A monocrystalline silicon wafer — a thin disc of up to 300 mm in diameter — serves as the substrate throughout this process. Each cycle begins with a chemical pre-treatment, such as the thermal growth of a thin oxide film for electrical isolation. A photosensitive material called *resist* is then deposited onto the wafer surface. An optical exposure system projects the pattern for the current layer — defined in a chromium film on a transparent quartz plate known as the *reticle*, or *photo-mask* — onto the resist. After development, the regions that received light are selectively dissolved (in the case of a positive resist), exposing the underlying material. This material then undergoes one of several process steps. *Plasma etching* selectively removes oxide or semiconductor material to define trenches and isolation regions. *Ion implantation* locally modifies the electrical properties of the silicon to form transistor junctions and doped wells. *Physical or chemical vapour deposition* lays down thin films of conductive metal — typically tungsten or copper — to form the interconnecting wiring that links the millions of functional elements stacked across all layers. After processing, the remaining resist is stripped and the cycle repeats for the next layer, until all layers are complete and the wafer is diced and packaged.

In the earliest days of the industry, the reticle was brought into direct physical contact with the resist-coated wafer during exposure — a technique known as *contact lithography*. While simple, this approach had two fundamental shortcomings. First, repeated physical contact progressively contaminated and damaged the delicate reticle pattern, necessitating its costly and time-consuming replacement. Second, a 1:1 projection offers no optical resolution advantage: any defect present on the reticle is transferred at full size onto the wafer. These limitations motivated a transition to *projection optical lithography*, in which a reduction lens images the reticle pattern onto the wafer from a safe working distance. The projection lens introduces a demagnification factor of four, meaning that features on the reticle are printed four times smaller on the wafer, immediately relaxing the demands on reticle fabrication. The minimum printable feature size, known as the *critical dimension* (CD), is governed by[13]:

$$\mathrm{CD} = k\,\frac{\lambda}{\mathrm{NA}}, \tag{1.0.1}$$

where λ is the illumination wavelength, NA is the numerical aperture of the projection lens, and k is a process-dependent factor. Reducing CD requires a shorter wavelength, a higher NA, and process improvements that lower k. This relation has driven a continuous push toward shorter wavelengths — from mercury arc-lamp sources at 436 nm and 365 nm, through excimer lasers at 248 nm and 193 nm, to the extreme ultraviolet (EUV) sources at 13.5 nm now entering production.

Achieving a high numerical aperture, however, comes with a practical constraint. Since $\mathrm{NA} = n \sin \vartheta$, where ϑ is the half-angle of the marginal ray accepted by the lens, attaining values of NA close to one in air requires the lens to subtend a very wide solid angle. Constructing a lens element that simultaneously covers the full area of a 300 mm wafer at high NA would require an optical element of impractical size and cost[13]. The industry's solution was to accept a smaller *exposure field* — typically 26×32 mm, corresponding to a single IC die — and expose the wafer in successive steps: after each die is exposed, the wafer stage repositions the wafer to the next die location and the exposure is repeated. This architecture defines the *wafer stepper*. In a stepper, each die is illuminated in a single, instantaneous full-field flash, with the stage brought to a complete stop before exposure commences.

While the stepper represented a decisive step forward, exposing an entire die simultaneously subjects the full field to any non-uniformities in the illumination intensity or lens aberrations. The *scanner* addresses this by replacing the full-field flash with a *scanning exposure*: the illumination system reveals only a narrow rectangular slit of the reticle, and the reticle stage sweeps the mask through this slit while the wafer stage moves synchronously in the opposite direction, maintaining the 4:1 demagnification relationship throughout[3;13]. Because the reticle moves at four times the speed of the wafer, it traverses the full reticle pattern in the same time that the wafer moves one die height. This scanning motion offers significant advantages over full-field exposure: it averages out illumination dose non-uniformities across the exposure field, improves imaging uniformity at the field edges, and allows the use of a physically larger reticle that is easier to write with electron-beam pattern generators. After each die is scanned, the wafer is stepped to the next die position and the scan repeats, defining the *step-and-scan* cycle that gives modern lithographic tools their name. A throughput of over 175 wafers per hour —

corresponding to less than 0.2 s of exposure time per die — is not uncommon in state-of-the-art tools[3].

A critical consequence of the multi-layer nature of IC fabrication is the requirement that each new layer be positioned with nanometer-level accuracy relative to all preceding layers — a specification known as *overlay*. The functional layers of a chip are interconnected by millions of tiny conductive pillars called *vias*; a misregistration between consecutive layers of even a few nanometers can misalign a via, break an intended connection, or create an unintended short circuit, rendering the IC defective[13]. Overlay requirements scale roughly in proportion with the CD and currently sit below 4 nm for leading-edge nodes. Overlay is therefore one of the primary technology drivers in scanner design, and it is governed directly by the positioning accuracy of the wafer and reticle stages during the scanning exposure.

The commercial performance of a scanner is ultimately measured in wafers per hour (WPH), since a machine whose capital cost can exceed $150 million must generate revenue continuously. Although the exposure time per die is fixed by the optics and the process, the time spent stepping and *settling* — the interval between the end of a step motion and the moment the stage has damped its residual vibrations to within the nanometer-level positioning specification — represents a significant and controllable fraction of the total cycle time[3], which places the allowable settle time below 10 ms. Every millisecond saved in settling directly translates into more dies exposed per hour and thus more chips delivered per unit time, with a lower cost per chip. Reducing settling time is therefore a central objective in scanner control engineering.

It is worth noting that, while EUV lithography at 13.5 nm has entered high-volume production for leading-edge nodes below 10 nm, deep ultraviolet (DUV) optical lithography at 193 nm remains the dominant technology for *mature nodes* — technology generations at 28 nm and above. Mature-node chips are the backbone of automotive electronics, industrial controllers, power management devices, analogue circuits, and the vast majority of consumer electronics.

The aggregate volume of wafers produced at mature nodes substantially exceeds that at leading-edge nodes, and is projected to remain so for the foreseeable future. Throughput improvements in DUV step-and-scan tools therefore continue to have a large global impact.

The key lever for reducing settling time — and thus for increasing scanner throughput — is the generation of *reference trajectories*: motion profiles that command the stage from its current position to the next scanning start-point with minimal excitation of mechanical resonances and the shortest possible residual settling tail. A well-designed trajectory limits the energy injected into the structural eigenmodes of the machine, reduces induced vibrations in the sensitive metrology and optical systems, and enables the stage to reach nanometer-level positioning accuracy well within the allocated window. The design of such trajectories is a problem of direct industrial relevance, and forms the central subject of this work.

1.1 Research Gap

Existing literature on trajectory design for wafer steppers evaluates candidate profiles exclusively through positioning metrics: tracking error, settling time, and the filtered exposure indices formalized in Chapter 3. What those metrics do not capture is the *morphology* of the failures into which these residual errors translate on the wafer. Yield engineering, meanwhile, has long exploited that same morphology in the opposite direction: the pattern of the failing dies across the wafer — its *fingerprint* — is routinely used to diagnose which process step caused an observed defect[10], and industrial datasets such as WM-811K[16] catalogue hundreds of thousands of labelled examples of these patterns. No published methodology runs this relationship forward: from a given set of high-order kinematic bounds, predict the defect morphology they will produce on the wafer, and use that prediction as a cost function to optimize the trajectory. The absence of this link means that more aggressive acceleration profiles cannot be explored in an informed way, simultaneously limiting the achievable throughput and the wafer yield.

1.2 Hypothesis

The central hypothesis of this work is:

> *The kinematic bounds ($v_{\max}$, $a_{\max}$, $j_{\max}$, $s_{\max}$) of a fourth-order reference trajectory can be optimized, subject to a throughput constraint, using as cost function the probability distribution over defect-morphology classes estimated by a convolutional neural network trained on industrial wafer fingerprints.*

For the hypothesis to be testable, we decompose into three falsifiable sub-claims tied to a measurable criterion:

- **H1 (Sensitivity)** The exposure-window synchronization error of the simulated machine responds monotonically and reproducibly to each active kinematic bound.
- **H2 (Expressiveness)** The synthetic wafer defect maps produced from the simulation can be meaningfully separated by the CNN such that only the scanner-plausible classes activate.
- **H3 (Optimality)** A population-based search over the four kinematic bounds, driven by the CNN cost, converges to profiles that overcome third-order and regular fourth-order baselines on the yield–throughput plane.

H1 is formulated because if sweeping a bound leaves exposure indices unchanged (or changes them erratically), it provides no gradient to effectively guide an optimizer. Similarly, **H2** is driven by the need to use the CNN probability distribution as a criterion to isolate the phenomenological causes behind the resulting error type on the wafer, while ensuring the simulation captures real scanner physics. Furthermore, **H3** establishes that a yield-optimal trajectory profile can be iteratively approximated to overcome conventional baseline limitations. Chapter 5 evaluates these three claims.

1.3 Objectives

The general objective is to develop a computational framework comprising a simulation of a lithography scanner and a convolutional neural network as a defect-evaluation model that can provide us with a yield–throughput plane over which a population-based optimizer can search for fourth-order trajectories that maximize estimated yield without penalizing throughput. This decomposes into six specific objectives:

O1 Implement the multi-body dynamic model of[2] as a simulation restricted to the x–y plane, capable of following a parameterized trajectory profile.

O2 Describe an analytical model to generate fourth-order trajectories from kinematic bounds (v_{max}, a_{max}, j_{max}, s_{max}) and segment times.

O3 Train and validate a convolutional neural network to classify the nine defect-morphology classes on WM-811K[16].

O4 Design a mapping from the dynamic error on a simulated traversal into a synthetic wafer defect map comparable with the CNN inputs.

O5 Implement a population-based optimizer coupled to the simulation using the CNN's probability distribution as a dynamic cost function to optimize the kinematic parameters of a reference trajectory.

O6 Evaluate the framework through comparative scenarios quantifying the throughput–yield trade-off against third-order and regular fourth-order baselines.

Objectives O1–O5 are implemented and validated in this document (Chapters 2–5), including the exact fifteen-segment snap-limited generator together with its second- and third-order counterparts (Section 4.3). O6 is addressed by the optimization scenarios E3 and E4 and by the profile-order comparison of Experiment 7.

1.4 Outline of the Thesis

Chapter 2 develops the formulation of the multi-body model of the machine in state-space. Chapter 3 translates the industrial drivers into quantitative control objectives. Chapter 4 describes the simulation environment, the trajectory generator, the stage controller, the CNN pipeline. Chapter 5 reports the experiments performed to validate the simulated machine, the evaluation of the classifier and the trajectory search. The conclusion summarizes findings, limitations, and further comparative scenarios.

Chapter 2

Model

2.1 Introduction

As we established before, the accurate relative positioning between the reticle and the wafer is crucial for *step-and-scan* machines to produce functional ICs. To analyze these positioning requirements in a multi-body setting, we leverage existing literature[2] that proposes modeling the lithography scanner as a combination of three open-loop serial chains: the reticle chain ($j = 1$, blue), the optics chain ($j = 2$, purple), and the wafer chain ($j = 3$, red); the latter being the crux of the problem studied here.

The base frame serves as the common structural root for all three chains. In the reticle chain, a dual-stage architecture is employed to satisfy the competing requirements for travel range and precision. Initially, the long-stroke motor (LS) —of the planar moving-coil type[14]— consisting of four coil actuators arranged in a platform shape facilitates large-range travel (of at least 300 mm) with moderate micrometer accuracy in 6-DOF. However, being directly mounted to the base frame, it is inherently susceptible to vibrations propagated from the rest of the machine.

To achieve nanometric tracking accuracy, a short-stroke motor (SS) is appended to the long-stroke carrier. Since travel in this stage is limited up to 1 mm, it is realized with Lorentz actuators which provide contact-free, high-bandwidth positioning. Given that the Lorentz force depends entirely on the current within the actuator and is independent of the stator's motion, it effectively decouples the end-effector from the base frame disturbances through magnetic suspension[3]. To minimize linearity errors and optimize force transfer, the long-stroke stage carrying the stator must maintain the optimal clearance to the magnets of the rest of the short-stroke actuator. This arrangement is replicated for the wafer chain.

The optics chain is attached to the base frame through a metrology frame which is in turn supported by pneumatic vibration isolators[3]. While the optics box includes adaptive optics with multiple elements, it is reduced in this framework to a single fixed lens. This allows the model to focus on machine compliance and structural vibration transfer rather than involving active optical control[2].

The physical connections between these stages, which constitute the primary transmission path for vibration, are represented by lumped-parameter spring-damper systems, according to the Kelvin-Voigt model. As its proponents state, this low-fidelity modeling choice is sufficient to study the dynamics between the multiple bodies during *step-and-scan* motion[2].

Figure 2.1 illustrates the structural topology and hierarchical arrangement within each chain. Horizontal Kelvin–Voigt elements $(K_{i,j}, C_{i,j})$ couple each stage laterally to the fixed structural reference depicted as a solid line in the color of the corresponding chain; in particular, elements (K_1, C_1) provide passive vibration isolation between $\mathcal{F}_0 \rightarrow \mathcal{F}_1$ through the solid green line that extends the base frame ($i = 1$) into the coupling to the first body ($i = 2$) in any j chain. The in-plane deviations $f_{i,j}$ represented by the horizontal blue double-headed arrows measure the relative displacement between each body. The dashed vertical centerline represents the nominal reference for $f^0_{i,j}$, when all relative coordinates are zero.

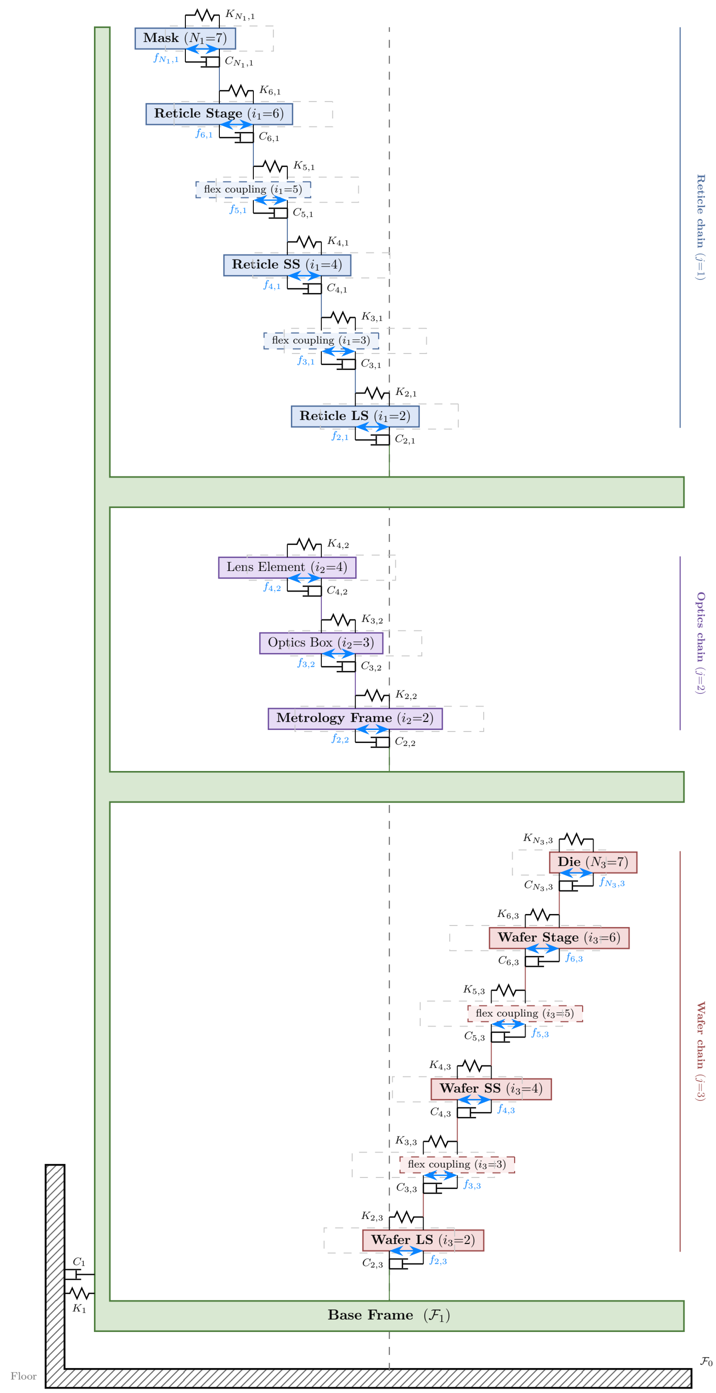


Figure 2.1: Lumped mass-spring-damper system of the three chains vertically stacked on the common base frame

2.2 Dynamics

Two distinct joint types arise: *actuated* stages ($i \in \{2, 4\}$), which are driven by servo motors and carry a near-infinite effective stiffness that keeps f_{i_j} locked to the nominal reference; and *flex* stages ($i \in \{3, 5, 6, 7\}$), which are passive compliant connections whose stiffness K_{i_j} and damping C_{i_j} are physical material properties of the mechanical coupling.

The full 6-DOF model is here restricted to in-plane motions[1;4], since rotations about the x and y axis, as well as translation on z are used mainly for focusing. The generalized coordinate used to describe the configuration of the ith stage within the jth chain is

$$q_{i_j} = \begin{bmatrix} f_{i_j} \\ \theta_{z_{i_j}} \end{bmatrix} \in \mathbb{R}^{3\times 1} \tag{2.2.1}$$

where $f_{i_j} = [f_{x,i_j},\ f_{y,i_j}]^T \in \mathbb{R}^2$ collects the in-plane translational displacements of stage i relative to its parent body in chain j, and $\theta_{z_{i_j}} \in \mathbb{R}$ is the corresponding rotation about the out-of-plane z-axis. The joints between bodies within a chain are modeled using a spring-damper system[2]. The associated generalized mass-inertia matrix for each body is

$$M_{i_j} = \begin{bmatrix} m_{i_j}\,\mathbf{I}_2 & 0 \\ 0 & I_{z_{i_j}} \end{bmatrix} \in \mathbb{R}^{3\times 3} \tag{2.2.2}$$

and the generalized stiffness and damping matrices $K_{i_j}, C_{i_j} \in \mathbb{R}^{3\times 3}$ are block-diagonal, with a 2×2 translational block and a scalar rotational entry.

The absolute position $f^0_{i_j}$ of body i is expressed as the recursive sum of its relative displacement from its parent body along the chain from the floor frame $\mathcal{F}_0$. Thus, for bodies with $i \geq 2$ on any chain $j \in \{1, 2, 3\}$:

$$f^0_{i_j} = f^0_{(i-1)_j} + f_{i_j} \tag{2.2.3}$$

Since the rotation about the z-axis is a scalar, the angular position recurrence is simply additive:

$$\theta^0_{z_{i_j}} = \theta^0_{z_{(i-1)_j}} + \theta_{z_{i_j}} \tag{2.2.4}$$

From this, the translational velocity expression appears as:

$$\dot{f}^0_{i_j} = \dot{f}^0_{(i-1)_j} + \dot{f}_{i_j} + \omega_{(i-1)_j} \times f_{i_j} \tag{2.2.5}$$

Where $\omega_{(i-1)_j}$ is the angular velocity[5] of $\mathcal{F}_{(i-1)_j}$ with respect to the floor, $\mathcal{F}_0$.

Taking the second derivative gives:

$$\begin{aligned}
\ddot{f}^0_{i_j} &= \ddot{f}^0_{(i-1)_j} + \ddot{f}_{i_j} + \omega_{(i-1)_j} \times \dot{f}_{i_j} + \dot{\omega_{(i-1)_j}} \times f_{i_j} + \omega_{(i-1)_j} \times \left[\dot{f}_{i_j} + \omega_{(i-1)_j} \times f_{i_j}\right] \\
&= \ddot{f}^0_{(i-1)_j} + \ddot{f}_{i_j} + \omega_{(i-1)_j} \times \dot{f}_{i_j} + \dot{\omega_{(i-1)_j}} \times f_{i_j} + \omega_{(i-1)_j} \times \dot{f}_{i_j} + \omega_{(i-1)_j} \times \left[\omega_{(i-1)_j} \times f_{i_j}\right] \\
&= \ddot{f}^0_{(i-1)_j} + \ddot{f}_{i_j} + 2\left[\omega_{(i-1)_j} \times \dot{f}_{i_j}\right] + \dot{\omega_{(i-1)_j}} \times f_{i_j} + \omega_{(i-1)_j} \times \left[\omega_{(i-1)_j} \times f_{i_j}\right]
\end{aligned} \tag{2.2.6}$$

Since we are concerned with planar movements in this work, we restrict the angular velocity of each body purely about the z-axis as $\omega = \dot{\theta}_z \, \hat{z}$, so that:

$$\omega_{(i-1)_j} = \dot{\theta}^0_{z_{(i-1)_j}} \, \hat{z}$$

We substitute the cross-products as matrix-vector products using the skew-symmetric matrix. In the 6-DOF model[2], it appears as:

$$[\omega] = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix}$$

Clearly, here we can set $\omega_x = 0$ and $\omega_y = 0$, thus $\omega = [0,\, 0,\, \dot{\theta}_z]^T$ and the matrix becomes:

$$[\omega] = \begin{bmatrix} 0 & -\omega_z & 0 \\ \omega_z & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Considering our 2D displacement vector, we only take the upper-left 2×2 sub-matrix, and factoring out the scalar $\dot{\theta}^0_{z_{(i-1)_j}}$:

$$[\omega_{(i-1)_j}] = \dot{\theta}^0_{z_{(i-1)_j}} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \tag{2.2.7}$$

such that $\omega_{(i-1)_j} \times f_{i_j} = [\omega_{(i-1)_j}]\, f_{i_j}$. Applying this matrix twice gives the centripetal term:

$$[\omega_{(i-1)_j}]^2 = -\left(\dot{\theta}^0_{z_{(i-1)_j}}\right)^2 \mathbf{I}_2 \tag{2.2.8}$$

so that $\omega_{(i-1)_j} \times [\omega_{(i-1)_j} \times f_{i_j}] = [\omega_{(i-1)_j}]^2 f_{i_j}$. Substituting and expanding (2.2.3) recursively from the floor frame, the translational kinematics become:

$$f^0_{i_j} = f^0_1 + \sum_{k=2}^{i} f_{k_j} \tag{2.2.9}$$

$$\dot{f}^0_{i_j} = \dot{f}^0_1 + \sum_{k=2}^{i} \dot{f}_{k_j} + \sum_{k=2}^{i} [\omega_{(k-1)_j}]\, f_{k_j} \tag{2.2.10}$$

$$\ddot{f}^0_{i_j} = \ddot{f}^0_1 + \sum_{k=2}^{i} \ddot{f}_{k_j} + \sum_{k=2}^{i} \left[2\,[\omega_{(k-1)_j}]\,\dot{f}_{k_j} + [\dot{\omega}_{(k-1)_j}]\, f_{k_j} + [\omega_{(k-1)_j}]^2 f_{k_j}\right] \tag{2.2.11}$$

Since the rotation $\theta_{z_{i_j}}$ is a scalar, no Coriolis or centripetal cross-terms arise in the angular kinematics. Expanding (2.2.4) recursively from the floor frame gives the rotational analogues:

$$\theta^0_{z_{i_j}} = \theta^0_{z_1} + \sum_{k=2}^{i} \theta_{z_{k_j}} \tag{2.2.12}$$

$$\dot{\theta}^0_{z_{i_j}} = \dot{\theta}^0_{z_1} + \sum_{k=2}^{i} \dot{\theta}_{z_{k_j}} \tag{2.2.13}$$

$$\ddot{\theta}^0_{z_{i_j}} = \ddot{\theta}^0_{z_1} + \sum_{k=2}^{i} \ddot{\theta}_{z_{k_j}} \tag{2.2.14}$$

Thus the equation of the angular motion of the ith body in the jth chain w.r.t $\mathcal{F}_0$ is:

$$^0I_j\,\dot{\omega}_j + ([\omega_j])\,^0I_j\,\omega_j = m^0_j \tag{2.2.15}$$

Under the planar restriction $\omega_x = \omega_y = 0$ the gyroscopic term vanishes like:

$$\left([\omega_{i_j}]\,^0I_{i_j}\right)\omega_{i_j} = \begin{bmatrix} 0 & -\dot{\theta}_z & 0 \\ \dot{\theta}_z & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} I_{xx} & 0 & 0 \\ 0 & I_{yy} & 0 \\ 0 & 0 & I_{zz} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_z \end{bmatrix} = 0$$

Leaving only:

$$I_{z_{i_j}}\,\ddot{\theta}^0_{z_{i_j}} = m^0_{z_{i_j}} \tag{2.2.16}$$

where $I_{z_{i_j}}$ is the aggregated moment of inertia of all bodies from i to the end-effector N_j, referred to the z-axis of body i via Steiner's theorem, and $m^0_{z_{i_j}} \in \mathbb{R}$ is the net applied moment acting on body i in chain j about the z-axis.

Steiner's theorem and inertia aggregation. The parallel axis (Steiner's) theorem states that the moment of inertia of a rigid body about any axis parallel to one through its center of mass (CoG) satisfies

$$I_z = \tilde{I}_z + m\,\|f\|^2, \tag{2.2.17}$$

where $\tilde{I}_z$ is the body's own moment of inertia about the z-axis through its CoG, m is its mass, and $\|f\|^2 = f_x^2 + f_y^2$ is the squared in-plane distance between the two parallel axes. In a serial chain, the inertia of every downstream body must be shifted to the frame of the current body using (2.2.17). Following the characterization article[2] (Algorithm 1 therein), the aggregated inertia for the jth chain is computed recursively from the end-effector inward. In the planar restriction, the full 3×3 inertia matrix of the original algorithm reduces to a scalar z-axis inertia:

Algorithm 1 Planar inertia aggregation for the jth chain — planar reduction of Algorithm 1 in[2]

Require: CoG inertias $\tilde{I}_{z_{i_j}}$, masses m_{i_j}, relative positions f_{k_j} for $i = 2, \ldots, N_j$

Ensure: Aggregated inertias ${}^0I_{z_{i_j}}$ for $i = 2, \ldots, N_j$

Initialize: ${}^0I_{z_{N_j}} \leftarrow \tilde{I}_{z_{N_j}}$
if $2 \le i < N_j$ **then**
  for $k = N_j,\ \ldots,\ i+1$ **do**
    $\tilde{m}_{(k-1)_j} \leftarrow \sum_{b=k}^{N_j} m_{b_j}$ ▷ accumulated downstream mass
    ${}^0I_{z_{(k-1)_j}} \leftarrow {}^0I_{z_{k_j}} + \tilde{I}_{z_{(k-1)_j}} + \tilde{m}_{(k-1)_j}\left(f_{x,k_j}^2 + f_{y,k_j}^2\right)$ ▷ Steiner shift from frame k to frame $k-1$
  end for
end if

Here $\tilde{m}_{(k-1)_j} = \sum_{b=k}^{N_j} m_{b_j}$ is the total mass of all downstream bodies (from k to the end-effector), $\tilde{I}_{z_{(k-1)_j}}$ is the local inertia of body $k-1$ about its own CoG, and $\tilde{m}_{(k-1)_j}(f_{x,k_j}^2 + f_{y,k_j}^2)$ is the Steiner correction that shifts the accumulated inertia from body k's CoG to body $(k-1)$'s CoG. By the time the algorithm terminates, ${}^0I_{z_{2_j}}$ contains the inertia of the entire chain j

referred to the CoG of its first body.

The aggregated inertia of the base frame (body 1) applies the same shift one more time, from body 2's CoG to the base frame:

$$
{}^{0}I_{z_1} = \tilde{I}_{z_1} + \sum_{j=1}^{3}\Big({}^{0}I_{z_{2_j}} + \tilde{m}_{1_j}\left(f_{x,2_j}^{2} + f_{y,2_j}^{2}\right)\Big), \tag{2.2.18}
$$

where $\tilde{m}_{1_j} = \sum_{b=2}^{N_j} m_{b_j}$ is the total mass of chain j. This is the quantity denoted I_{z_0} in (2.2.22).

Expanding $\ddot{\theta}^{0}_{z_{i_j}}$ using (2.2.14):

$$
I_{z_{i_j}}\ddot{\theta}^{0}_{z_1} + I_{z_{i_j}}\sum_{k=2}^{i}\ddot{\theta}_{z_{k_j}} = m^{0}_{z_{i_j}} \tag{2.2.19}
$$

2.2.1 For the base frame

To write the equations that govern the base frame, we start from Newton's equation $F = ma$. For every body ($i \geq 2$) within chain j, it connects to body $i-1$ through a spring above and body $i+1$ through the spring below:

$$
\begin{aligned}
m_{i_j}\ddot{f}^{0}_{1} + m_{i_j}\sum_{k=2}^{i}\ddot{f}_{k_j} + m_{i_j}\sum_{k=2}^{i}\Big[2\left[\omega_{(k-1)_j}\right]\dot{f}_{k_j} + \left[\dot{\omega}_{(k-1)_j}\right]f_{k_j} + \left[\omega_{(k-1)_j}\right]^{2}f_{k_j}\Big] \\
-K_i\left(f^{0}_{(i-1)_j} - f^{0}_{i_j}\right) - C_i\left(\dot{f}^{0}_{(i-1)_j} - \dot{f}^{0}_{i_j}\right) + K_{i+1}\left(f^{0}_{i_j} - f^{0}_{(i+1)_j}\right) + C_{i+1}\left(\dot{f}^{0}_{i_j} - \dot{f}^{0}_{(i+1)_j}\right) = u^{0}_{i_j}
\end{aligned} \tag{2.2.20}
$$

Summing (2.2.20) over all bodies in a chain (taking the optics chain $j = 2$, bodies $i = 2, 3, 4$ as a representative example), the internal spring forces cancel by Newton's third law:

$$
\begin{aligned}
&- K_2(f^{0}_{1} - f^{0}_{2}) - C_2(\dot{f}^{0}_{1} - \dot{f}^{0}_{2}) \\
&\cancelto{0}{+K_3(f^{0}_{2} - f^{0}_{3}) + C_3(\dot{f}^{0}_{2} - \dot{f}^{0}_{3}) - K_3(f^{0}_{2} - f^{0}_{3}) - C_3(\dot{f}^{0}_{2} - \dot{f}^{0}_{3})} \\
&\cancelto{0}{+K_4(f^{0}_{3} - f^{0}_{4}) + C_4(\dot{f}^{0}_{3} - \dot{f}^{0}_{4}) - K_4(f^{0}_{3} - f^{0}_{4}) - C_4(\dot{f}^{0}_{3} - \dot{f}^{0}_{4})}
\end{aligned}
$$

Only the boundary term $-K_2(f^{0}_{1} - f^{0}_{2}) - C_2(\dot{f}^{0}_{1} - \dot{f}^{0}_{2})$ survives, which corresponds to the connection between body $i = 2$ and the base frame. This extends to any N_j-body chain. Recalling (2.2.3), $f_1^0 - f_{2_j}^0 = -f_{2_j}$, and following[2] with ${}^0K_{i_j} = {}^0R_{i_j} K_{i_j} {}^0R_{i_j}^T$ and ${}^0C_{i_j} = {}^0R_{i_j} C_{i_j} {}^0R_{i_j}^T$, the total translational equation of the base frame (absorbing the balance masses and collecting all chain contributions) is:

$$m_0\, \ddot{f}_1^0 + {}^0\bar{C}_1\, \dot{f}_1 + {}^0\bar{K}_1\, f_1 - \sum_{j=1}^{3} \left\{ {}^0\hat{C}_{2_j}\, \dot{f}_{2_j} + {}^0\hat{K}_{2_j}\, f_{2_j} \right\} = u_1^0 \tag{2.2.21}$$

where $m_0 = m_1 + \sum_{j=1}^{3} \left(m_B^j + \sum_{i=2}^{N_j} m_{i_j} \right)$ is the total system mass, and the stiffness and damping matrices denote the 2×2 translational block. Following[2], ${}^0\bar{K}_1 = {}^0K_1$ and ${}^0\bar{C}_1 = {}^0C_1$, while the hatted chain couplings are ${}^0\hat{C}_{2_j} = {}^0C_{2_j}$ and ${}^0\hat{K}_{2_j} = {}^0K_{2_j} + {}^0C_{2_j}[\omega_1]$; the accents are defined in general in (2.2.29) below.

By the exact same cancellation argument applied to the moment balance about the z-axis — the moment exerted by the spring connecting body i to body $i-1$ is $-K_{i_j}\, \theta_{z_{i_j}}$, so internal moments cancel identically — the rotational equation of the base frame is:

$$I_{z_0}\, \ddot{\theta}_{z_1}^0 + {}^0\bar{C}_1\, \dot{\theta}_{z_1} + {}^0\bar{K}_1\, \theta_{z_1} - \sum_{j=1}^{3} \left\{ {}^0\hat{C}_{2_j}\, \dot{\theta}_{z_{2_j}} + {}^0\hat{K}_{2_j}\, \theta_{z_{2_j}} \right\} = \tau_1^0 \tag{2.2.22}$$

where $I_{z_0} = I_{z_1} + \sum_{j=1}^{3} \left(I_{z_B}^j + \sum_{i=2}^{N_j} I_{z_{i_j}} \right)$ is the total moment of inertia of the system about the z-axis, ${}^0\bar{C}_{i_j}$ and ${}^0\bar{K}_{i_j}$ (and their hatted counterparts, which coincide with them for the scalar rotational entries) are the scalar rotational damping and stiffness of the joint connecting body i to its predecessor in chain j, and τ_1^0 is the net applied torque on the base frame.

2.2.2 For each chain

Recalling (2.2.20) and (2.2.19), for the first body in the chain ($i = 2$), which connects to the base frame above and to body $i = 3$ below, the translational equation is:

$$\bar{m}_{2_j} \ddot{f}_{2_j}^0 + {}^0\bar{C}_{2_j}\, \dot{f}_{2_j} + {}^0\bar{K}_{2_j}\, f_{2_j} - \left\{ {}^0\hat{C}_{3_j}\, \dot{f}_{3_j} + {}^0\hat{K}_{3_j}\, f_{3_j} \right\} = u_{2_j}^0 \tag{2.2.23}$$

and the rotational equation is:

$$I_{z_{2_j}} \ddot{\theta}^0_{z_{2_j}} + {}^0\bar{C}_{2_j}\, \dot{\theta}_{z_{2_j}} + {}^0\bar{K}_{2_j}\, \theta_{z_{2_j}} - {}^0\hat{C}_{3_j}\, \dot{\theta}_{z_{3_j}} - {}^0\hat{K}_{3_j}\, \theta_{z_{3_j}} = \tau^0_{2_j} \tag{2.2.24}$$

For bodies $3 \leq i < N_j$, the same structure holds with bodies $2, \ldots, i-1$ in between. The translational equation is:

$$\bar{m}_{i_j} \ddot{f}^0_{i_j} + {}^0\bar{C}_{i_j}\, \dot{f}_{i_j} + {}^0\bar{K}_{i_j}\, f_{i_j} - \left\{ {}^0\hat{C}_{(i+1)_j}\, \dot{f}_{(i+1)_j} + {}^0\hat{K}_{(i+1)_j}\, f_{(i+1)_j} \right\} = u^0_{i_j} \tag{2.2.25}$$

and the rotational equation is:

$$I_{z_{i_j}} \ddot{\theta}^0_{z_{i_j}} + {}^0\bar{C}_{i_j}\, \dot{\theta}_{z_{i_j}} + {}^0\bar{K}_{i_j}\, \theta_{z_{i_j}} - {}^0\hat{C}_{(i+1)_j}\, \dot{\theta}_{z_{(i+1)_j}} - {}^0\hat{K}_{(i+1)_j}\, \theta_{z_{(i+1)_j}} = \tau^0_{i_j} \tag{2.2.26}$$

Finally, for body $i = N_j$ (the end effector), there is no subsequent body in the chain. The translational and rotational equations are:

$$\bar{m}_{N_j j} \ddot{f}^0_{N_j j} + {}^0\bar{C}_{N_j j}\, \dot{f}_{N_j j} + {}^0\bar{K}_{N_j j}\, f_{N_j j} = 0 \tag{2.2.27}$$

$$I_{z_{N_j j}} \ddot{\theta}^0_{z_{N_j j}} + {}^0\bar{C}_{N_j j}\, \dot{\theta}_{z_{N_j j}} + {}^0\bar{K}_{N_j j}\, \theta_{z_{N_j j}} = 0 \tag{2.2.28}$$

2.2.3 Accented matrices and the nominal condition

Following Appendix A of[2], the accented matrices appearing in the equations above are not merely rotated joint matrices: they absorb the rotational cross-terms of (2.2.11) into effective damping and stiffness. With $[\omega]$ the planar skew form of (2.2.7) and $B_{i_j} = [\dot{\omega}_{i_j}] + [\omega_{i_j}]^2$:

$$\begin{aligned}
\bar{m}_{i_j} &= \sum_{k=i}^{N_j} m_{k_j} \\
{}^0\bar{C}_{i_j} &= {}^0C_{i_j} + 2m_{i_j}\,[\omega_{(i-1)_j}] \\
{}^0\bar{K}_{i_j} &= {}^0K_{i_j} + {}^0C_{i_j}\,[\omega_{(i-1)_j}] + m_{i_j}\, B_{(i-1)_j} \\
{}^0\hat{C}_{(i+1)_j} &= {}^0C_{(i+1)_j} \\
{}^0\hat{K}_{(i+1)_j} &= {}^0K_{(i+1)_j} + {}^0C_{(i+1)_j}\,[\omega_{i_j}]
\end{aligned} \tag{2.2.29}$$

The bar thus marks the matrices acting on the body's *own* joint coordinates, which collect the Coriolis ($2m[\omega]$), damper-convection ($C[\omega]$), and Euler–centripetal (mB) contributions, while the hat marks the coupling to the *downstream* joint coordinates, which inherits only the convective damper term: the damper reacts to the inertial velocity $\dot{f} + [\omega]f$ of (2.2.10), so part of its force is proportional to position and migrates into the effective stiffness. The scalar rotational entries admit no such cross-terms in the planar restriction, so about the z-axis the accents are inert and $\hat{(\cdot)} = \bar{(\cdot)} = (\cdot)$. For $3 \leq i < N_j$ the model of[2] additionally couples body i to every upstream joint through the two-index matrices ${}^0\hat{C}_{i_j,k} = 2m_{i_j}\,[\omega_{(k-1)_j}]$ and ${}^0\hat{K}_{i_j,k} = m_{i_j}\,B_{(k-1)_j}$ for $2 \leq k < i$, omitted above.

Every one of these corrections is proportional to $\dot{\theta}_z$ or $\ddot{\theta}_z$ of a parent body. At the nominal operating condition $\dot{\theta}_z = \ddot{\theta}_z = 0$ — the regulated step-and-scan regime studied throughout this work — they vanish identically: the accented matrices reduce to the frame-rotated ${}^0K_{i_j}$ and ${}^0C_{i_j}$, the inertial accelerations of (2.2.11) reduce to sums of relative accelerations, and the equations of motion coincide with Eqs. (2.6)–(2.8) of[2] and become linear time-invariant. Away from the nominal condition, the corrections re-enter as the parameter-varying part $A_1(\Omega, \dot{\Omega})$ of the decomposition given in[2]. Note that the scheduling variables Ω are themselves state variables — the $\dot{\theta}_z$ of each body — so the cross-terms are quadratic in the state (and the rotated matrices ${}^0K_{i_j}$, ${}^0C_{i_j}$ carry a further state dependence through ${}^0R_{i_j}(\theta_z)$): the machine model is, strictly, nonlinear, a *quasi*-LPV system whose frozen nominal dynamics are the LTI model above. The MuJoCo test bed used throughout this work assembles the full nonlinear multibody dynamics from this lumped topology directly, retaining these couplings without approximation; the LTI form above is recovered only as its frozen nominal dynamics at $\dot{\theta}_z = \ddot{\theta}_z = 0$, not as a separately integrated reduced model.

2.2.4 State-space formulation

Because each body possesses three planar degrees of freedom (f_x, f_y, θ_z) along with their corresponding time derivatives, it contributes exactly six states to the system. Following the machine description of[2] (Section 2.2 therein), the complete machine model comprises 6 bodies in the reticle chain ($N_1 = 7$), 3 bodies in the optics chain ($N_2 = 4$), and 6 bodies in the wafer chain ($N_3 = 7$). Including the common base frame (body 1), the system features a total of $1 + (N_1 - 1) + (N_2 - 1) + (N_3 - 1) = 16$ bodies. Consequently, the global state vector x encompasses $6 \times 16 = 96$ states:

$$x = \Big[\underbrace{x_1, \ldots, x_6}_{\text{base frame}}, \quad \underbrace{x_7, \ldots, x_{6N_1}}_{\text{chain 1: bodies } 2 \to N_1}, \quad \underbrace{\cdots}_{\text{chain 2}}, \quad \underbrace{\cdots}_{\text{chain 3}}\Big]^T \tag{2.2.30}$$

The explicit mapping for each state is detailed in the table below. Here, p denotes the global sequential index of the i-th body in the j-th chain within the concatenated list of all bodies (starting with the base frame, followed sequentially by chains 1, 2, and 3). It is crucial to note that every positional entry within the state vector represents a *relative* joint displacement; the corresponding absolute inertial position $f^0_{i_j}$ can be reconstructed using the recursive summations from (2.2.9) to (2.2.12).

Solving (2.2.21) and (2.2.22) for the base-frame accelerations yields:

$$\ddot{f}^0_1 = -\frac{{}^0\bar{C}_1}{m_0}\,\dot{f}_1 - \frac{{}^0\bar{K}_1}{m_0}\,f_1 + \frac{1}{m_0}\sum_{j=1}^{3}\left\{{}^0\bar{C}_{2_j}\,\dot{f}_{2_j} + {}^0\bar{K}_{2_j}\,f_{2_j}\right\} + \frac{u^0_1}{m_0} \tag{2.2.31}$$

$$\ddot{\theta}^0_{z_1} = -\frac{{}^0\bar{C}_1}{I_{z_0}}\,\dot{\theta}_{z_1} - \frac{{}^0\bar{K}_1}{I_{z_0}}\,\theta_{z_1} + \frac{1}{I_{z_0}}\sum_{j=1}^{3}\left\{{}^0\bar{C}_{2_j}\,\dot{\theta}_{z_{2_j}} + {}^0\bar{K}_{2_j}\,\theta_{z_{2_j}}\right\} + \frac{\tau^0_1}{I_{z_0}} \tag{2.2.32}$$

Table 2.2.1: State-space mapping for the global concatenated system vector

State-space Variable	Expression	Meaning
x_1	f_{x_1}	base frame x-position
x_2	f_{y_1}	base frame y-position
x_3	θ_{z_1}	base frame rotation
x_4	$\dot{f}_{x_1}$	base frame x-velocity
x_5	$\dot{f}_{y_1}$	base frame y-velocity
x_6	$\dot{\theta}_{z_1}$	base frame angular velocity
x_7	$f_{x_{2_1}}$	chain 1, body 2, x-position
x_8	$f_{y_{2_1}}$	chain 1, body 2, y-position
x_9	$\theta_{z_{2_1}}$	chain 1, body 2, rotation
x_{10}	$\dot{f}_{x_{2_1}}$	chain 1, body 2, x-velocity
x_{11}	$\dot{f}_{y_{2_1}}$	chain 1, body 2, y-velocity
x_{12}	$\dot{\theta}_{z_{2_1}}$	chain 1, body 2, angular velocity
$\vdots$	$\vdots$	$\vdots$
x_{6p-5}	$f_{x_{i_j}}$	chain j, body i, x-position
x_{6p-4}	$f_{y_{i_j}}$	chain j, body i, y-position
x_{6p-3}	$\theta_{z_{i_j}}$	chain j, body i, rotation
x_{6p-2}	$\dot{f}_{x_{i_j}}$	chain j, body i, x-velocity
x_{6p-1}	$\dot{f}_{y_{i_j}}$	chain j, body i, y-velocity
x_{6p}	$\dot{\theta}_{z_{i_j}}$	chain j, body i, angular velocity
$\vdots$	$\vdots$	$\vdots$

Expressed in the standard matrix form $\dot{x} = Ax + Bu$, the first six rows governing the base-frame dynamics are:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \\ \dot{x}_5 \\ \dot{x}_6 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & \cdots \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & \cdots \\ -\frac{K_{1,xx}}{m_0} & -\frac{K_{1,xy}}{m_0} & 0 & -\frac{C_{1,xx}}{m_0} & -\frac{C_{1,xy}}{m_0} & 0 & \frac{K_{2_1,xx}}{m_0} & \frac{K_{2_1,xy}}{m_0} & 0 & \frac{C_{2_1,xx}}{m_0} & \cdots \\ -\frac{K_{1,yx}}{m_0} & -\frac{K_{1,yy}}{m_0} & 0 & -\frac{C_{1,yx}}{m_0} & -\frac{C_{1,yy}}{m_0} & 0 & \frac{K_{2_1,yx}}{m_0} & \frac{K_{2_1,yy}}{m_0} & 0 & \frac{C_{2_1,yx}}{m_0} & \cdots \\ 0 & 0 & -\frac{K_{1,zz}}{I_{z_0}} & 0 & 0 & -\frac{C_{1,zz}}{I_{z_0}} & 0 & 0 & \frac{K_{2_1,zz}}{I_{z_0}} & 0 & \cdots \end{bmatrix} x + \begin{bmatrix} 0 & 0 & 0 & \cdots \\ 0 & 0 & 0 & \cdots \\ 0 & 0 & 0 & \cdots \\ \frac{1}{m_0} & 0 & 0 & \cdots \\ 0 & \frac{1}{m_0} & 0 & \cdots \\ 0 & 0 & \frac{1}{I_{z_0}} & \cdots \end{bmatrix} \begin{bmatrix} u^0_{x,1} \\ u^0_{y,1} \\ \tau^0_1 \\ \vdots \end{bmatrix} \tag{2.2.33}$$

General State-Space Form for the Chain Bodies

To systematically construct the global state-space matrices, we must reframe the equations of motion for the chain bodies in terms of the state vector variables. We map the physical planar coordinates of the p-th sequential body to the local state vectors: translational position $\mathbf{x}_{2p-1} = [x_{6p-5}, x_{6p-4}]^T$, translational velocity $\dot{\mathbf{x}}_{2p-1} = [x_{6p-2}, x_{6p-1}]^T$, rotational position $\mathbf{x}_{2p} = x_{6p-3}$, and rotational velocity $\dot{\mathbf{x}}_{2p} = x_{6p}$.

The state derivatives are dictated by the relative accelerations between adjacent bodies. By applying the kinematic identity $\ddot{f}_{i_j} = \ddot{f}^0_{i_j} - \ddot{f}^0_{(i-1)_j}$ from (2.2.11), we determine the interactive forces acting on each body within its parent's reference frame. Let p represent the global index of body i in chain j, while $p-1$ and $p+1$ denote the indices of its immediate upstream and downstream neighbors respectively (with the base frame positioned at $p = 1$).

For the **first body in each chain** ($i = 2$, index p, upstream index 1), the translational state derivatives ($\ddot{\mathbf{x}}_{2p-1} = \ddot{f}_{2_j}$) and rotational state derivatives ($\ddot{\mathbf{x}}_{2p} = \ddot{\theta}_{z_{2_j}}$) take the form:

$$\begin{aligned}
\ddot{\mathbf{x}}_{2p-1} &= \frac{1}{\bar{m}_{2_j}}\Big[u^0_{2_j} - {}^0\bar{C}_{2_j}\dot{\mathbf{x}}_{2p-1} - {}^0\bar{K}_{2_j}\mathbf{x}_{2p-1} + {}^0\hat{C}_{3_j}\dot{\mathbf{x}}_{2p+1} + {}^0\hat{K}_{3_j}\mathbf{x}_{2p+1}\Big] \\
&\quad - \frac{1}{m_0}\Big[u^0_1 - {}^0\bar{C}_1\dot{\mathbf{x}}_1 - {}^0\bar{K}_1\mathbf{x}_1 + \sum_{k=1}^{3}\Big({}^0\hat{C}_{2_k}\dot{\mathbf{x}}_{2p_{2_k}-1} + {}^0\hat{K}_{2_k}\mathbf{x}_{2p_{2_k}-1}\Big)\Big]
\end{aligned} \tag{2.2.34}$$

$$\begin{aligned}
\ddot{\mathbf{x}}_{2p} &= \frac{1}{I_{z_{2_j}}}\Big[\tau^0_{2_j} - {}^0\bar{c}_{\theta,2_j}\dot{\mathbf{x}}_{2p} - {}^0\bar{k}_{\theta,2_j}\mathbf{x}_{2p} + {}^0\hat{c}_{\theta,3_j}\dot{\mathbf{x}}_{2p+2} + {}^0\hat{k}_{\theta,3_j}\mathbf{x}_{2p+2}\Big] \\
&\quad - \frac{1}{I_{z_0}}\Big[\tau^0_1 - {}^0\bar{C}_1\dot{\mathbf{x}}_2 - {}^0\bar{K}_1\mathbf{x}_2 + \sum_{k=1}^{3}\Big({}^0\hat{c}_{\theta,2_k}\dot{\mathbf{x}}_{2p_{2_k}} + {}^0\hat{k}_{\theta,2_k}\mathbf{x}_{2p_{2_k}}\Big)\Big]
\end{aligned} \tag{2.2.35}$$

For a **general intermediate body** ($2 < i < N_j$, index p), the dynamics are shaped by upstream, local, and downstream interactions:

$$\begin{aligned}
\ddot{\mathbf{x}}_{2p-1} &= \frac{1}{\bar{m}_{i_j}}\Big[u^0_{i_j} - {}^0\bar{C}_{i_j}\dot{\mathbf{x}}_{2p-1} - {}^0\bar{K}_{i_j}\mathbf{x}_{2p-1} + {}^0\hat{C}_{(i+1)_j}\dot{\mathbf{x}}_{2p+1} + {}^0\hat{K}_{(i+1)_j}\mathbf{x}_{2p+1}\Big] \\
&\quad - \frac{1}{\bar{m}_{(i-1)_j}}\Big[u^0_{(i-1)_j} - {}^0\bar{C}_{(i-1)_j}\dot{\mathbf{x}}_{2p-3} - {}^0\bar{K}_{(i-1)_j}\mathbf{x}_{2p-3} + {}^0\hat{C}_{i_j}\dot{\mathbf{x}}_{2p-1} + {}^0\hat{K}_{i_j}\mathbf{x}_{2p-1}\Big]
\end{aligned} \tag{2.2.36}$$

$$\begin{aligned}
\ddot{\mathbf{x}}_{2p} &= \frac{1}{I_{z_{i_j}}}\Big[\tau^0_{i_j} - {}^0\bar{c}_{\theta,i_j}\dot{\mathbf{x}}_{2p} - {}^0\bar{k}_{\theta,i_j}\mathbf{x}_{2p} + {}^0\hat{c}_{\theta,(i+1)_j}\dot{\mathbf{x}}_{2p+2} + {}^0\hat{k}_{\theta,(i+1)_j}\mathbf{x}_{2p+2}\Big] \\
&\quad - \frac{1}{I_{z_{(i-1)_j}}}\Big[\tau^0_{(i-1)_j} - {}^0\bar{c}_{\theta,(i-1)_j}\dot{\mathbf{x}}_{2p-2} - {}^0\bar{k}_{\theta,(i-1)_j}\mathbf{x}_{2p-2} + {}^0\hat{c}_{\theta,i_j}\dot{\mathbf{x}}_{2p} + {}^0\hat{k}_{\theta,i_j}\mathbf{x}_{2p}\Big]
\end{aligned} \tag{2.2.37}$$

Finally, for the **end-effector body** ($i = N_j$, index p), the downstream force terms naturally vanish as there are no subsequent bodies:

$$\begin{aligned}
\ddot{\mathbf{x}}_{2p-1} &= \frac{1}{\bar{m}_{N_{j_j}}}\Big[u^0_{N_{j_j}} - {}^0\bar{C}_{N_{j_j}}\dot{\mathbf{x}}_{2p-1} - {}^0\bar{K}_{N_{j_j}}\mathbf{x}_{2p-1}\Big] \\
&\quad - \frac{1}{\bar{m}_{(N_j-1)_j}}\Big[u^0_{(N_j-1)_j} - {}^0\bar{C}_{(N_j-1)_j}\dot{\mathbf{x}}_{2p-3} - {}^0\bar{K}_{(N_j-1)_j}\mathbf{x}_{2p-3} + {}^0\hat{C}_{N_{j_j}}\dot{\mathbf{x}}_{2p-1} + {}^0\hat{K}_{N_{j_j}}\mathbf{x}_{2p-1}\Big]
\end{aligned} \tag{2.2.38}$$

$$\begin{aligned}
\ddot{\mathbf{x}}_{2p} &= \frac{1}{I_{z_{N_{j_j}}}}\Big[\tau^0_{N_{j_j}} - {}^0\bar{c}_{\theta,N_{j_j}}\dot{\mathbf{x}}_{2p} - {}^0\bar{k}_{\theta,N_{j_j}}\mathbf{x}_{2p}\Big] \\
&\quad - \frac{1}{I_{z_{(N_j-1)_j}}}\Big[\tau^0_{(N_j-1)_j} - {}^0\bar{c}_{\theta,(N_j-1)_j}\dot{\mathbf{x}}_{2p-2} - {}^0\bar{k}_{\theta,(N_j-1)_j}\mathbf{x}_{2p-2} + {}^0\hat{c}_{\theta,N_{j_j}}\dot{\mathbf{x}}_{2p} + {}^0\hat{k}_{\theta,N_{j_j}}\mathbf{x}_{2p}\Big]
\end{aligned} \tag{2.2.39}$$

Chapter 3

Control Objectives

3.1 Introduction

In Chapter 1 we identified overlay and throughput as the two primary drivers of scanner design, followed in Chapter 2 by the modeling of its multi-body dynamics. This chapter focuses on mapping those industrial requirements into quantitative objectives that any candidate reference trajectory must satisfy. Here we explore in detail how the synchronization error arises between the wafer and the reticle stages, examine the filtered metrics through which the industry evaluates it, and establish the formal statement of the optimization problem.

3.2 Synchronization Error

During the scanning exposure, the projection optics image the reticle onto the wafer with a demagnification factor of $\alpha = 0.25$. Thus, the image of the reticle pattern is stationary on the wafer surface only if the wafer follows the reticle motion scaled by α at every instant. The degradation of the printed pattern is therefore not due to the individual tracking error of either stage, but their scaled difference. Let $r_w(t)$ and $r_r(t)$ denote the reference positions of the wafer and reticle stages respectively along the scanning direction, and $y_w(t)$, $y_r(t)$ their measured positions. The *synchronization error* is defined as[3]

$$e_{syn}(t) = \big(y_w(t) - r_w(t)\big) - \alpha\big(y_r(t) - r_r(t)\big). \tag{3.2.1}$$

Naturally, a common-mode mode disturbance that displaces both stages consistently with the projection ratio leaves e_{syn} unaffected. Conversely, any differential motion between the chains — exacerbated by the lightly damped structural resonance — is transferred directly to the printed image.

3.3 Exposure Metrics: MA and MSD

For an area of photoresist to become exposed, the illumination dose must overcome a threshold. Despite direct exposure during the scanning motion, the photoresist on the wafer does not respond to the instantaneous value of e_{syn} but to its aggregate effect over the time a given point stays below the light source. Therefore, industry practice evaluates the synchronization error through two moving-window filters matched to the exposure time t_{exp} of one point on the wafer[3]:

- First, the **moving average**

$$\mathrm{MA}(t_k) = \frac{1}{N}\sum_{n=0}^{N-1} e_{syn}(t_{k-n}) \tag{3.3.1}$$

  measures the mean displacement of the image and manifests as *overlay* error or the misalignment of the printed pattern with respect to the previous layer.

- Second, the **moving standard deviation**

$$\mathrm{MSD}(t_k) = \sqrt{\frac{1}{N}\sum_{n=0}^{N-1}\Big(e_{syn}(t_{k-n}) - \mathrm{MA}(t_k)\Big)^2} \tag{3.3.2}$$

  measures the smearing of the aerial image around that mean position and manifests as *fading*, a loss of contrast that distorts the printed feature.

For the 38 nm half-pitch process, the specifications adopted[2] are

$$\mathrm{MA} \in [-1.25,\ +2]\ \mathrm{nm}, \qquad \mathrm{MSD} \leq 7\ \mathrm{nm}, \tag{3.3.3}$$

evaluated at every instant during the exposure window[3]. These bounds must be met while the stage sustains the required scanning velocity, operating within a settling time budget of under 10 ms to preserve wafer throughput. On the two-stage machine modelled here they are not a fixed pass/fail but the tight end of a trade curve: Chapter 5 shows that a sufficiently smooth fourth-order trajectory meets them — both MSD and MA, to the nanometer — at the cost of a longer scan, so the specification and the throughput budget are the two ends of the optimization this thesis performs.

3.4 Problem Statement

The reference trajectories studied here are fourth-order profiles parameterized by the kinematic bounds ($v_{\max}$, $a_{\max}$, $j_{\max}$, $s_{\max}$) on velocity, acceleration, jerk, and snap respectively. These four variables set both the duration of the motion — and hence throughput — and the frequency content of the excitation injected into the machine structure, and thereby the synchronization error during exposure.

Conventional trajectory design evaluates candidates using positioning metrics like settling time, peak error, or the MA/MSD bounds of (3.3.3). Such metrics make it possible to predict which dies will be defective, but reveal nothing about the *morphology* of the failures that accumulate across the wafer. Wafer fabs routinely exploit the converse relationship to diagnose which process step caused a particular spatial pattern of failing dies (the *fingerprint*)[10]. However, the forward direction remains an open problem as no published methodology predicts, from a given set of trajectory parameters, the defect pattern they will produce, nor uses that prediction as the cost function during trajectory optimization. This thesis addresses precisely that forward direction.

To evaluate the impact of a candidate trajectory on the overall process yield, the resulting wafer defect map is classified by a convolutional neural network trained on the WM811K dataset[16]. The network estimates a probability P_c for each defect morphology class c, and the trajectory cost is defined as the weighted sum of these probabilities:

$$\mathcal{J} = \sum_{c \in \mathcal{C}} w_c P_c, \tag{3.4.1}$$

where w_c is the economic weight of class c[8;9;7] and $\mathcal{C}$ is restricted to the classes plausibly caused by scanner motion (namely *none*, *Scratch*, *Edge-Loc*, and *Random*). The remaining classes (*Center*, *Donut*, *Edge-Ring*, *Loc*, *Near-full*) originate in subsequent steps of the manufacturing process. If they were to be activated, it would indicate a modeling error rather than a trajectory effect; therefore, this restriction is used as a validation criterion for the trajectory-to-defect mapping in Chapter 5.

The optimization problem is then formulated as

$$\min_{(v_{\max},\, a_{\max},\, j_{\max},\, s_{\max})} \mathcal{J} \qquad \text{subject to} \qquad T_{wafer} \leq T_{budget}, \tag{3.4.2}$$

where T_{wafer} is the total time to traverse a wafer determined by the trajectory parameters, and T_{budget} is the cycle-time budget corresponding to the target throughput. The central hypothesis of this work is that (3.4.2) can be solved using a data-driven cost function of the form (3.4.1), producing trajectory parameters that satisfy throughput constraints while minimizing scanner-induced defect costs.

Chapter 4

Methodology

4.1 Introduction

Trajectory generation is a well-studied problem. Trajectories with smooth higher-order derivatives are known to reduce the excitation of poorly damped structural modes. To evaluate the yield impact of our proposed trajectories without requiring physical scanner hardware, we need a simulation testbed where they can be executed and analyzed.

4.2 Simulation

In this section we describe the implementation of the dynamics model[2] described in Chapter 2 using the MuJoCo[15] physics engine for Python. The simulation serves as a testbed for validating the proposed trajectories and evaluating the synthetic wafer defect maps generated by their traversal.

The MuJoCo[15] Python bindings allow us to define the context of our simulation from an XML string by calling the `mujoco.MjModel.from_xml_string(MODEL_XML)` method. We extend from MuJoCo's `<worldbody>` tag on which we place a root `<body>` tag, representing our *base* *frame*; this allows us to organize the machine in the simulation using the three chain hierarchical tree. Table 4.2.1 maps the specific names used in the MuJoCo XML to their counterparts in the structural diagram of Figure 2.1.

Table 4.2.1: Mapping of MuJoCo XML bodies to the model components

`<body>` `name` property	Model body
`base_frame`	Base Frame ($\mathcal{F}_1$)
`w_ls_act`	Wafer LS ($i_3 = 2$)
`w_ls_flex`	flex coupling ($i_3 = 3$)
`w_ss_act`	Wafer SS ($i_3 = 4$)
`w_ss_flex`	flex coupling ($i_3 = 5$)
`w_stage`	Wafer Stage, passive ($i_3 = 6$)
`wafer`	Die ($N_3 = 7$)
`r_ls_act`	Reticle LS ($i_1 = 2$)
`r_ls_flex`	flex coupling ($i_1 = 3$)
`r_ss_act`	Reticle SS ($i_1 = 4$)
`r_ss_flex`	flex coupling ($i_1 = 5$)
`r_stage`	Reticle Stage, passive ($i_1 = 6$)
`mask`	Mask ($N_1 = 7$)
`metro_frame`	Metrology Frame ($i_2 = 2$)
`optics_box`	Optics Box ($i_2 = 3$)
`lens`	Lens Element ($i_2 = 4$)

The simulation environment is defined through several specialized XML attributes that handle different aspects of the physical world. Table 4.2.2 summarizes the function of each tag within the context of our lithography scanner model.

The mechanical properties of the couplings, represented by the Kelvin-Voigt elements, are

Table 4.2.2: MuJoCo XML tag functions in the scanner simulation

Tag	Function
`<worldbody>`	Defines the inertial root, the floor plane ($\mathcal{F}_0$).
`<body>`	Creates a rigid body with its own local coordinate system.
`<inertial>`	Assigns mass m_{i_j} and the aggregated inertia tensor I_{i_j}.
`<joint>`	Implements the degrees of freedom f_{i_j} and $\theta_{z_{i_j}}$.
`<geom>`	Defines the material (granite, silicon, glass) and collision shapes.
`<actuator>`	Models the Lorentz and moving-coil motors via servo controllers.

implemented via the `stiffness` and `damping` attributes of the `<joint>` tags. To begin, the common base frame ($\mathcal{F}_1$) is anchored to the world frame ($\mathcal{F}_0$) via a 3-DOF joint (two prismatic, one revolute) representing the isolation system with stiffness K_1 and damping C_1, as seen in Eq. (2.2.21) and (2.2.22). For the two actuated stages of each chain (long-stroke and short-stroke), where the physical stiffness is replaced by the controller's active response, the simulation uses high K_P values (`kp` $= 10^7$ N/m, to approximate the stiffness of a rigid servo motor while maintaining numerical stability with the implicit integrator at a timestep of $dt = 10^{-4}$ s).

Tables 4.2.3, 4.2.4, and 4.2.5 illustrate the distribution of these variables for the three chains.

To successfully adapt our planar reduction of the reference model[2] into MuJoCo's multibody solver, two numerical adaptations are in order:

1. Native joint damping against absolute velocity would exert artificial drag ($K_V/K_P =$ 10 mm per m/s) during cruise, so we apply the velocity gain K_V to the tracking-error velocity ($\dot{r} - \dot{y}$) within the control loop to represent the frictionless air bearings.

2. Actuators transmit reaction forces to the parent body by default, so we apply reaction cancellation to the base frame at each integration step to represent balance masses used to absorb long-stroke reaction forces[3], which the analytical model[2] represents as external

Table 4.2.3: Wafer chain stiffness (K) and damping (C) parameters

Coupling Interface	Stiffness Variable (K)	Damping Variable (C)
Floor → Base Frame	`K1`	`C1`
Base → LS Stage	`KP_ACT`	`KV_ACT_TRANS`
LS Stage → LS Flex	`K_LS_F`	`C_LS_F`
LS Flex → SS Stage	`KP_ACT`	`KV_ACT_TRANS`
SS Stage → SS Flex	`K_SS_F`	`C_SS_F`
SS Flex → Stage	`K_ST_F`	`C_ST_F`
Stage → Die	`K8_W`	`C8_W`

Table 4.2.4: Reticle chain stiffness (K) and damping (C) parameters

Coupling Interface	Stiffness Variable (K)	Damping Variable (C)
Floor → Base Frame	`K1`	`C1`
Base → LS Stage	`KP_ACT`	`KV_ACT_TRANS`
LS Stage → LS Flex	`K_LS_F`	`C_LS_F`
LS Flex → SS Stage	`KP_ACT`	`KV_ACT_TRANS`
SS Stage → SS Flex	`K_SS_F`	`C_SS_F`
SS Flex → Stage	`K_ST_F`	`C_ST_F`
Stage → Mask	`K8_R`	`C8_R`

Table 4.2.5: Optics chain stiffness (K) and damping (C) parameters

Coupling Interface	Stiffness Variable (K)	Damping Variable (C)
Base Frame $\rightarrow$ Metro Frame	`K2_0`	`C2_0`
Metro Frame $\rightarrow$ Optics Box	`K3_0`	`C3_0`
Optics Box $\rightarrow$ Lens Element	`K4_0`	`C4_0`

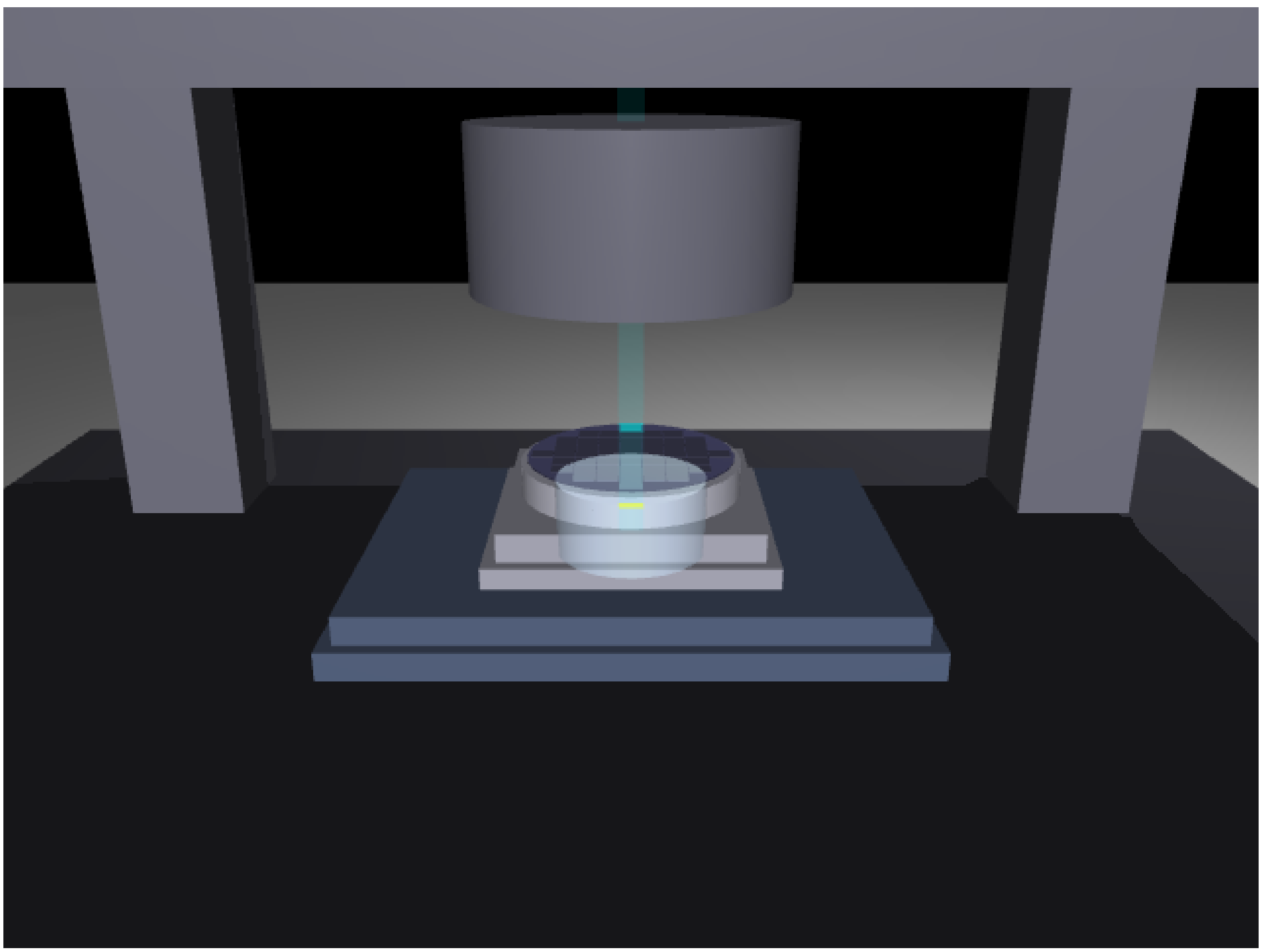


Figure 4.1: Screenshot of the MuJoCo simulation in the midst of the scanning motion, showing the wafer and the optics chain under the metrology frame

stage forces.

Synchronized scanning motion is commanded with nominal cruise velocities of $v_{w,\text{scan}} = 0.8\,\text{m/s}$ and $v_{r,\text{scan}} = 3.2\,\text{m/s}$. Under acceleration transients up to $a_{w,\max} = 20\,\text{m/s}^2$, this setup captures the full-state dynamic interactions derived in Section 2.2.

Baseline control configurations (such as Case 1 in[2]) are realized on this plant by disabling active fine-stage feedback loops.

4.3 Reference Trajectory Generation

The trajectories commanded to the stages are **fourth-order profiles**: piecewise-polynomial motions in which position, velocity, acceleration, and jerk are all continuous, and the fourth derivative — the *snap* — switches between $\pm s_{\max}$ and zero[11]. Within each segment the position evolves as

$$p(t) = p_0 + v_0 t + \frac{a_0}{2}t^2 + \frac{j_0}{6}t^3 + \frac{s}{24}t^4, \tag{4.3.1}$$

where $(p_0,\ v_0,\ a_0,\ j_0)$ are the boundary conditions inherited from the previous segment and $s \in \{-s_{\max}, 0, +s_{\max}\}$. A symmetric point-to-point motion consists of fifteen such segments, arranged so that none of the kinematic bounds $(v_{\max},\ a_{\max},\ j_{\max},\ s_{\max})$ is exceeded.

We implement an analytical trajectory generator that structures the motion profile across fifteen segments (seven per acceleration ramp) where snap is held constant. Physical limitations are addressed by the following considerations from[11]:

- When the jerk plateau cannot be reached, the effective jerk reduces to $\sqrt{a_{\max} s_{\max}}$
- If the acceleration limit is unreachable, then the effective acceleration follows from the corresponding quadratic equation; if the jerk phase collapses as well, the snap-phase duration instead follows from a cube-root solve

- If the total travel distance is too short to accommodate the required transition ramps, then the cruise velocity is iteratively reduced via bisection.

To streamline the trials, this same algorithmic structure is used to derive the lower order profiles: i.e., a second-order, acceleration-bounded trapezoidal profile and a third-order, seven-segment jerk-bounded S-curve.

The order of the motion profile determines the high-frequency content of the force commanded to initiate such motion: a second-order profile demands instantaneous jumps in acceleration, while a third-order profile is discontinuous in jerk[4], so both inject energy across a broad frequency band. A fourth-order trajectory, whose snap is bounded as noted above, is reported to roll off the excitation spectrum at $-80\,\mathrm{dB}$ per decade[11] — a theoretical asymptotic for constant-snap profiles, not separately verified spectrally in this work — reducing the response of poorly damped structural modes such as those of the end-effectors, the reticle and the wafer. The settling period after each step[11;1] is thereby minimized, which is essential for the constant-velocity scanning phase, where ideally no force beyond the viscous terms is present.

4.4 Stage Feedback Control

The stage controllers are evaluated across four incremental configurations, following[2]. Each setup introduces a specific feedforward or feedback component to isolate its impact on exposure performance in Chapter 5.

In Case 1, each long-stroke stage is regulated in closed loop by a collocated PD controller acting on the long-stroke encoder $y_c(t)$:

$$u(t) = K_P\, e(t) + K_V\, \dot{e}(t), \tag{4.4.1}$$

where $e(t) = r(t) - y_c(t)$ is the long-stroke tracking error, with gains $K_P = 10^7\,\mathrm{N/m}$ and

$K_V = 10^5$ N·s/m. Position and velocity are read from the long-stroke stage rather than the end-effector because non-collocated control through the lightly damped reticle coupling ($\zeta \approx 0.04$) destabilizes the loop. Consequently, the lag between the long-stroke stage and the end-effector remains uncompensated and appears directly in the synchronization error. On this frictionless plant, this lag is dominated by its acceleration-proportional component: servo deflection $m\,a/K_P$ and quasi-static coupling stretch $m_d\,a/K$ carrying the downstream mass m_d. This configuration serves as the baseline validated in Experiment 1 (Section 5.2).

Since both lag components are functions of the known reference kinematics, in Case 2 they are cancelled by commanding the long stroke *ahead* of the reference:

$$r_{LS}(t) = r(t) - \big(\ell_v\,\dot{r}(t) + \ell_a\,\ddot{r}(t)\big), \tag{4.4.2}$$

so that the end-effector may land on the desired position. The coefficients $(\ell_v,\,\ell_a)$ for each chain are identified by performing iterative least-squares regressions on the end-effector lag with respect to reference velocity and acceleration. Each resulting fit is added to the feedforward control loop while subsequent passes repeat the regression on the remaining residuals. Three iterations are required to prevent post-acceleration coupling vibrations from skewing the quasi-static fit.This is a model-free equivalent to the spring-compensation configuration of the reference study[2].

The error remaining after feedforward is a non-repeatable transient response that can only be suppressed through feedback. Thus, in Case 3, the short-stroke stage of each chain implements a closed position loop on the end-effector error $e_f(t)$, commanding the short-stroke actuator directly. In actual step-and-scan systems, this measurement is readily available from interferometry, as required for exposure metrology. Case 3a uses a linear PI law with anti-windup, whereas Case 3b replaces the linear integrator with a Hybrid Integrator–Gain System (HIGS)[6], given by:

$$\dot{x}_1 = \omega_h\,e_f \quad \text{while } e_f\,x_1 \geq x_1^2/k_h, \qquad x_1 = k_h\,e_f \quad \text{otherwise}, \tag{4.4.3}$$

with output $u_f = k_p e_f + \omega_i \int x_1\,dt$. By resetting its state to the gain line $k_h e_f$ whenever the sector condition is violated, the HIGS element provides integral action with roughly 38° less phase lag than a linear integrator, relaxing the overshoot-versus-stiffness limitation of linear designs — the property that motivates nonlinear integrators for wafer stages. The loop gains are tuned empirically over a grid up to the stability boundary of the simulated plant: the PI loop uses $k_p = 16$, $k_i = 480\,\mathrm{s}^{-1}$, and the HIGS loop $k_p = 16$, $\omega_i = 2\pi \cdot 200\,\mathrm{rad/s}$, $\omega_h = 2\pi \cdot 80\,\mathrm{rad/s}$, $k_h = 1$; one step further in gain destabilizes both loops against the 180 Hz flexible-chain modes. The HIGS element sustains an integrator frequency roughly 42× above the PI's equivalent k_i/k_p before reaching that boundary — the phase advantage of the nonlinear integrator expressed as usable gain. The fine-stage authority is ultimately bounded by the short-stroke actuator's own velocity loop (pole at $K_P/K_V = 100\,\mathrm{rad/s}$), a structural property of this generic replication that Chapter 5 quantifies.

4.5 Population-Based Trajectory Search

The optimization problem (3.4.2) is explored with particle swarm optimization (PSO) over the four kinematic bounds. Each particle holds a position $\mathbf{x} = (v_{\max}, a_{\max}, j_{\max}, s_{\max})$ and is updated by the standard rule

$$\mathbf{u}_{k+1} = \omega\,\mathbf{u}_k + c_1 r_1(\mathbf{p}_k - \mathbf{x}_k) + c_2 r_2(\mathbf{g}_k - \mathbf{x}_k), \qquad \mathbf{x}_{k+1} = \mathbf{x}_k + \mathbf{u}_{k+1}, \tag{4.5.1}$$

with inertia $\omega = 0.6$, cognitive and social weights $c_1 = c_2 = 1.4$, velocities clamped to ± 0.5 of the search-box span per dimension, and positions clipped to the search box. Evaluating one particle runs the full step-and-scan sequence over the wafer under the candidate trajectory, converts the per-die exposure indices into a wafer map, and prices the map. Two cost definitions are compared as the scenarios E3 and E4 of the evaluation plan: a conventional servo cost (the wafer-average exposure MSD, normalized by the failure threshold) and the yield-aware cost built on the CNN. For the latter, the weighted form (3.4.1) is instantiated as

$$\mathcal{J}_{map} = \big(1 - P_{none}\big) + \sum_{c\,\in\,\mathcal{C}_{ood}} P_c, \tag{4.5.2}$$

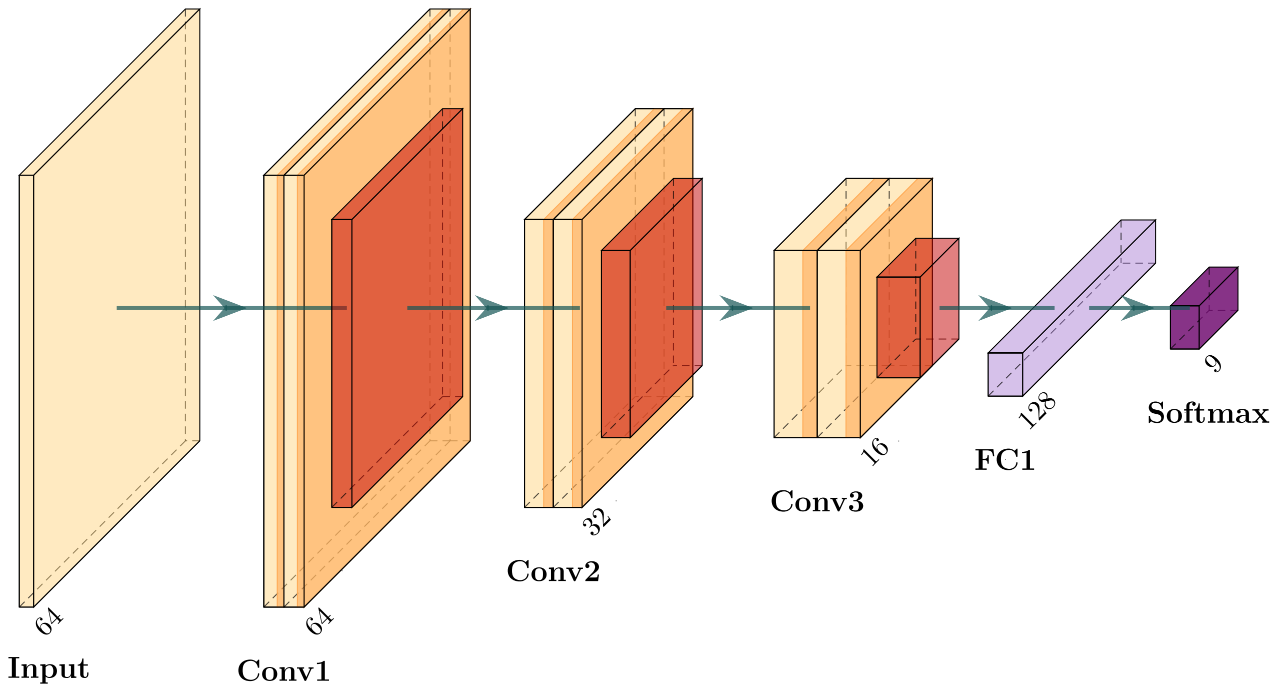
where $\mathcal{C}_{ood}$ is the set of classes that scanner motion cannot plausibly produce. The first term prices any departure from a clean map, so that the optimizer can spend an *acceptable-pattern* yield loss when it is worth it; the second is a hard wall that keeps it from cheapening the wafer into a morphology the classifier attributes to non-scanner mechanisms. Both scenarios share a throughput reward $\mathcal{J} = \mathcal{J}_{map} + \gamma\, T_{wafer}/T_{ref}$ with $\gamma = 1$ and $T_{ref} = 15\,\mathrm{s}$, so that a faster wafer is cheaper and the search trades cycle time against the map cost rather than merely respecting a budget; T_{wafer} follows from the scan-profile duration, the settling budget, and the die count.

4.6 From Synchronization Error to Wafer Maps

The link between trajectory parameters and yield is established in three steps: the simulated synchronization error is aggregated per die, the resulting pass/fail pattern forms a simulated wafer map, and a convolutional neural network classifies that map into a defect-morphology class.

The CNN classifier is trained on WM-811K[16], a public dataset of 811,457 wafer maps collected from real fabrication lines, of which 172,950 carry a manually assigned failure label. Each map is a grid of dies marked as pass or fail, annotated with one of nine morphology classes: *none*, *Center*, *Donut*, *Edge-Loc*, *Edge-Ring*, *Loc*, *Near-full*, *Random*, and *Scratch*. The labelled maps are split 80/20 with class stratification, and every map is resized to 64×64 cells to give the network a uniform input.

The network architecture (Figure 4.2) comprises three convolutional blocks (16, 32, and 64 filters of size 3×3, each followed by ReLU activation and 2×2 max-pooling) and two fully connected layers (128 hidden units with dropout of 0.5, and a 9-way softmax output). It is trained for 5 epochs with the Adam optimizer (learning rate 10^{-3}, batch size 32) under a cross-entropy loss, retaining the weights with the highest hold-out accuracy.



Layer	**Type**	**Operation**	**Shape**	**Activation**
Input Map	Wafer Grid	—	$1 \times 64 \times 64$	—
Conv1	2D Convolution	16 filters, 3×3, pad 1	$16 \times 64 \times 64$	ReLU
Pool1	Max Pooling	2×2 pool, stride 2	$16 \times 32 \times 32$	—
Conv2	2D Convolution	32 filters, 3×3, pad 1	$32 \times 32 \times 32$	ReLU
Pool2	Max Pooling	2×2 pool, stride 2	$32 \times 16 \times 16$	—
Conv3	2D Convolution	64 filters, 3×3, pad 1	$64 \times 16 \times 16$	ReLU
Pool3	Max Pooling	2×2 pool, stride 2	$64 \times 8 \times 8$	Flatten (4096)
FC1	Dense	Linear $(4096 \to 128)$	128	ReLU + Dropout ($p = 0.5$)
Output	Dense	Linear $(128 \to 9)$	9	Softmax (9 classes)

Figure 4.2: Architecture of the CNN (top) and detailed layer specification (bottom).

To generate synthetic wafer maps, the synchronization error recorded during each die exposure is reduced to the exposure-window MA/MSD metrics of Chapter 3 and evaluated against the threshold of Eq. (3.3.3) to declare each die as passing (1) or failing (2). The resulting per-die status values are interleaved with their spatial coordinates, producing an $N \times N$ trinary matrix W_{raw} (with 0 for background) of resolution inversely proportional to die dimension L_{die}.

The variable-resolution wafer defect map is scaled to a uniform 64×64 grid using OpenCV's `cv2.resize`. Since values in the grid represent discrete categorical labels, nearest-neighbor interpolation (`cv2.INTER_NEAREST`) is employed rather than linear or bicubic interpolation. The resulting $64 \times 64 \times 1$ tensor is presented to the CNN, which returns the probability distribution P_c used by the cost function (3.4.1).

Furthermore, the CNN serves as validation of the underlying multi-body model. For the trajectory-to-defect transformation to be meaningful, only the defect classes that could be caused by scanner motion must be activated: *none*, *Scratch*, *Edge-Loc*, and *Random*. The remaining classes correspond to downstream processes like etching, cleaning, and deposition; if the maps from the simulation activated them systematically, the mapping would be inaccurate.

Chapter 5

Experiments and Results

5.1 Introduction

At last, this chapter summarizes the experiments carried out on the framework described in the previous Chapter 4.

5.2 Experiment 1: Reproduction of the Case 1 Study

The first experiment reproduces Case 1 of the reference study by Al-Rawashdeh et al.[2]. Following the in-plane (x, y, θ_z) planar formulation established in Chapter 2, step-and-scan motion of the wafer and reticle chains is evaluated without the short-stroke stages. Under this baseline configuration, motion is governed solely by the long-stroke stage using the collocated PD controller of Section 4.4; the integral component is omitted to prevent windup during high-acceleration ramps. We aim to verify that the simulation reproduces the qualitative dynamic behavior reported in the literature, and to quantify how far a long-stroke-only machine falls short of the exposure specifications.

The step-and-scan sequence is performed over four dies ($32 \times 32\,\mathrm{mm}^2$ each in a single row) with baseline trajectory parameters $v_{w,scan} = 0.8\,\mathrm{m/s}$, $a_{w,\max} = 20\,\mathrm{m/s^2}$, $j_{\max} = 1600\,\mathrm{m/s^3}$, and $s_{\max} = 10^5\,\mathrm{m/s^4}$. The scan profile lasts 160.8 ms, with the first exposure window falling at $[53.6, 107.2]$ ms into the run, and the complete hold-plus-four-die sequence spans 1.843 s. The reticle moves at 3.2 m/s with $a_{r,\max} = 80\,\mathrm{m/s^2}$ in the opposite direction. Figure 5.1 shows the commanded references, and Figure 5.2 illustrates the resulting tracking and synchronization errors with the exposure windows highlighted.

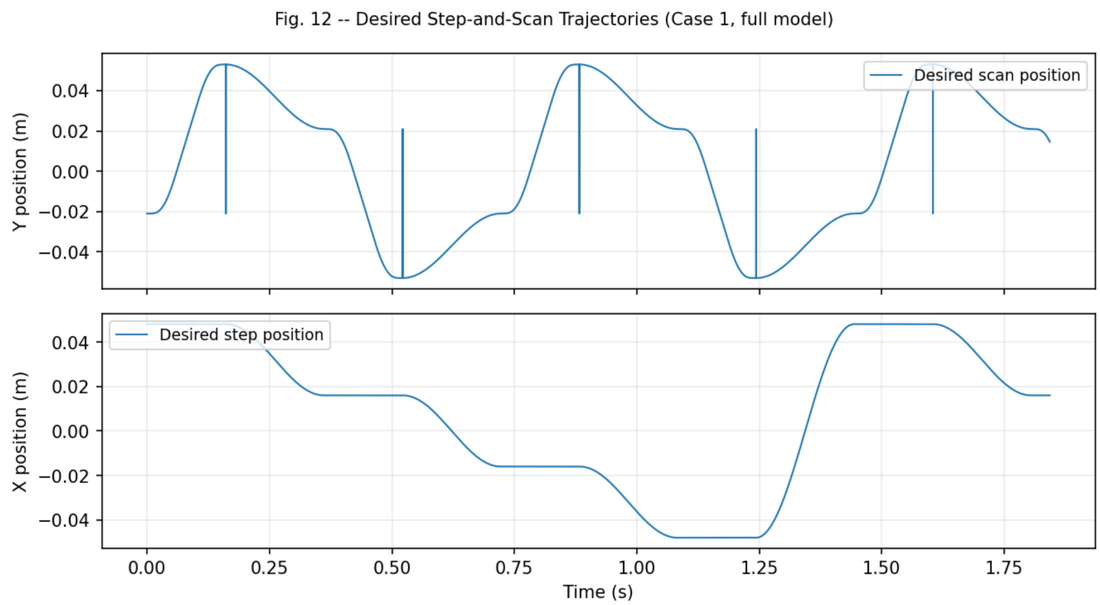


Figure 5.1: References trajectories for the wafer stage (x: stepping, y: scanning).

Closed-loop feedback is read exclusively at the long-stroke encoder (Section 4.4); the rest of the chain — comprising the short-stroke stage, the passive Stage element ($m_7 = 10.5\,\mathrm{kg}$, 180.0 Hz resonance), and the end-effector — remains unactuated. During transition ramps, this uncompensated flexibility accumulates acceleration-proportional lag, which, in combination with the reticle chain's higher speed (3.2 m/s) and softer coupling, produces the stage-to-end-effector coupling lag recorded in Figure 5.3.

Figure 5.4 shows both indices over the sequence, and Figure 5.5 provides the detail of the first exposure window. During the exposure (cruise) window, the performance indices evaluate to:

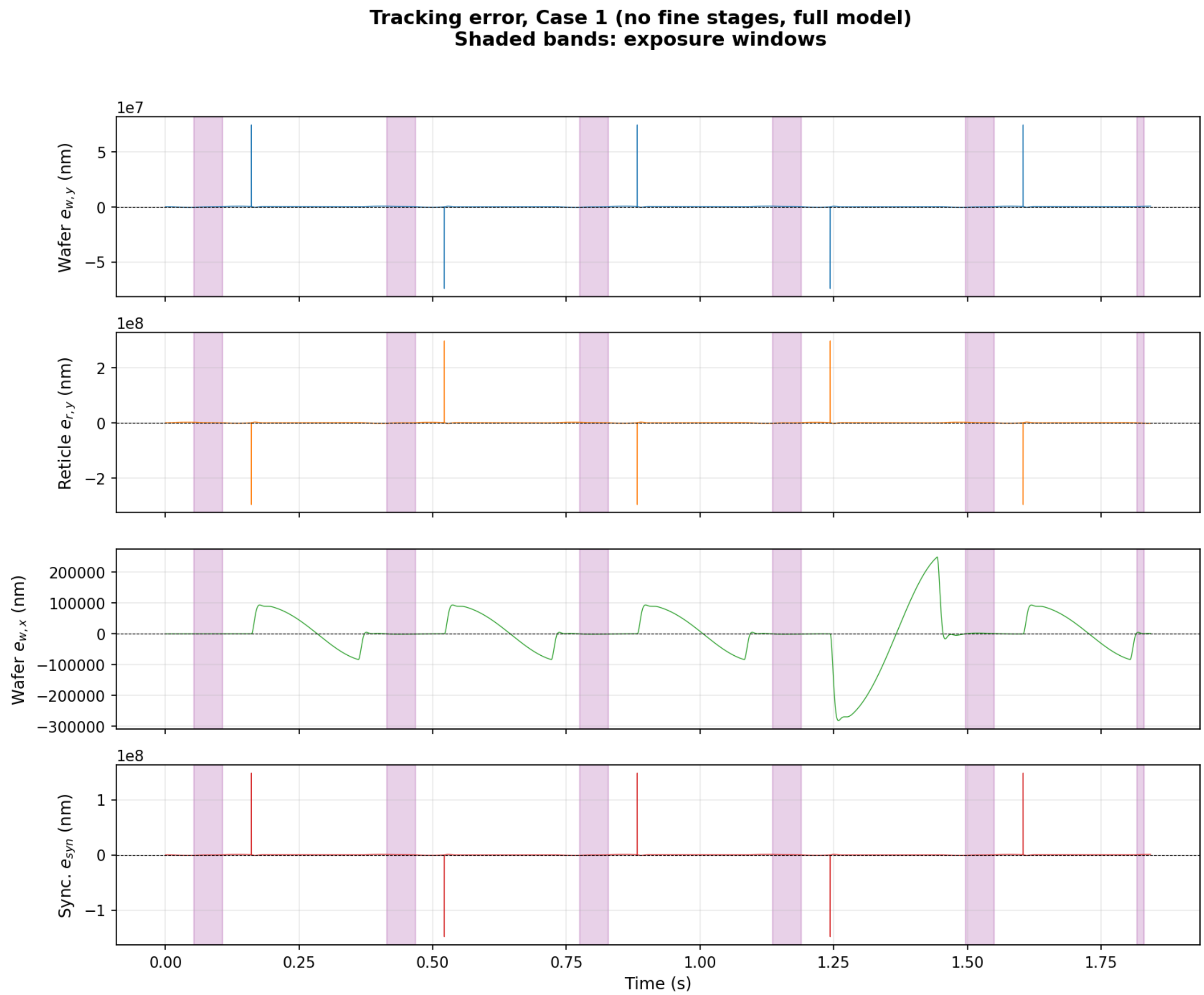


Figure 5.2: Wafer, reticle, and synchronization errors. Shaded bands mark the exposure windows.

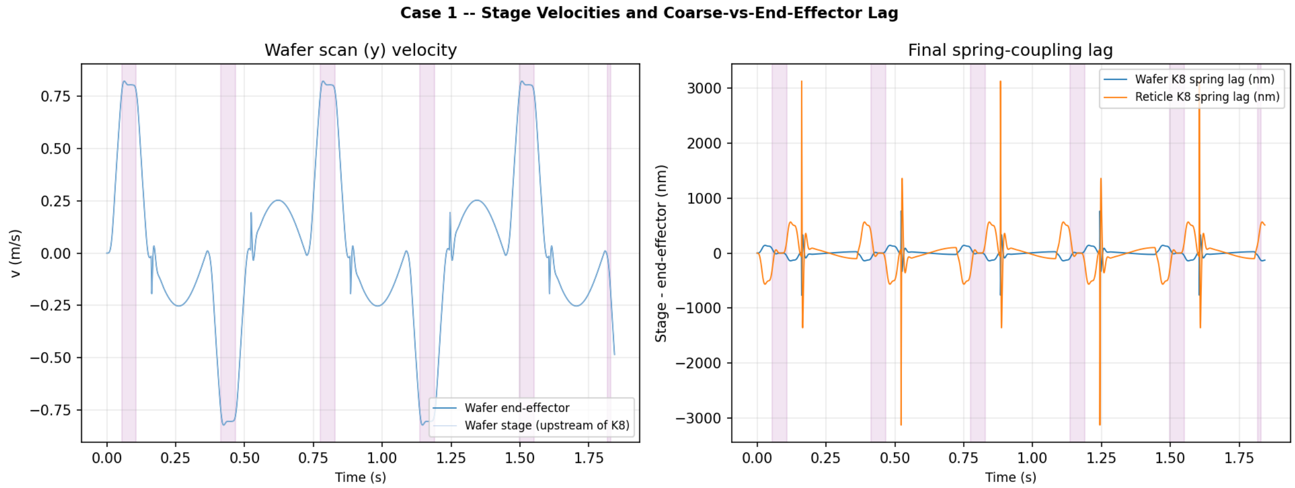


Figure 5.3: Stage velocities and spring-coupling lag during the scan.

$$\max|\mathrm{MA}_{syn}| = 954{,}955 \text{ nm} \approx 955.0\,\mu\text{m}, \qquad \max\mathrm{MSD}_{syn} = 25{,}034 \text{ nm} \approx 25.0\,\mu\text{m}, \qquad (5.2.1)$$

As in the reference study[2], a machine operating under long-stroke control alone misses the exposure specification by several orders of magnitude ($\max|\mathrm{MA}_{syn}|/\mathrm{MA}_{spec} \approx 4.8 \times 10^5$, $\max\mathrm{MSD}_{syn}/\mathrm{MSD}_{spec} \approx 3.6 \times 10^3$), motivating either fine-stage compensation downstream or gentler kinematic bounds.

5.3 Experiment 2: CNN Training and Evaluation on WM-811K

This experiment consists of the training and evaluation of the Convolutional Neural Network (Section 4.6). From the overall WM-811K dataset, the 172,950 labeled maps are partitioned using a stratified 80/20 split, leaving 34,590 maps for evaluation.

The network achieves an overall accuracy of **96.88 %** on the test set with a macro-averaged F1 score of 0.77. Table 5.3.1 reports the per-class precision, recall, and F1, and Figure 5.6 shows

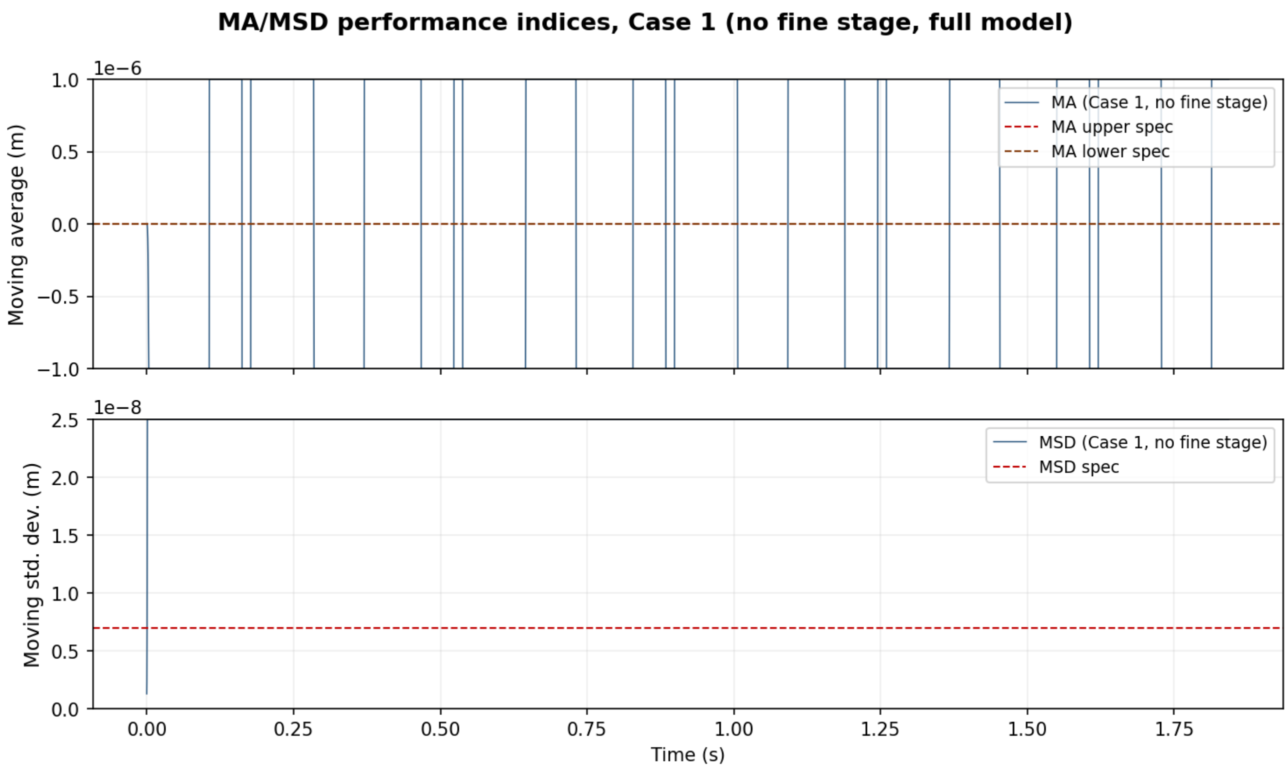


Figure 5.4: MA and MSD performance indices of the synchronization error.

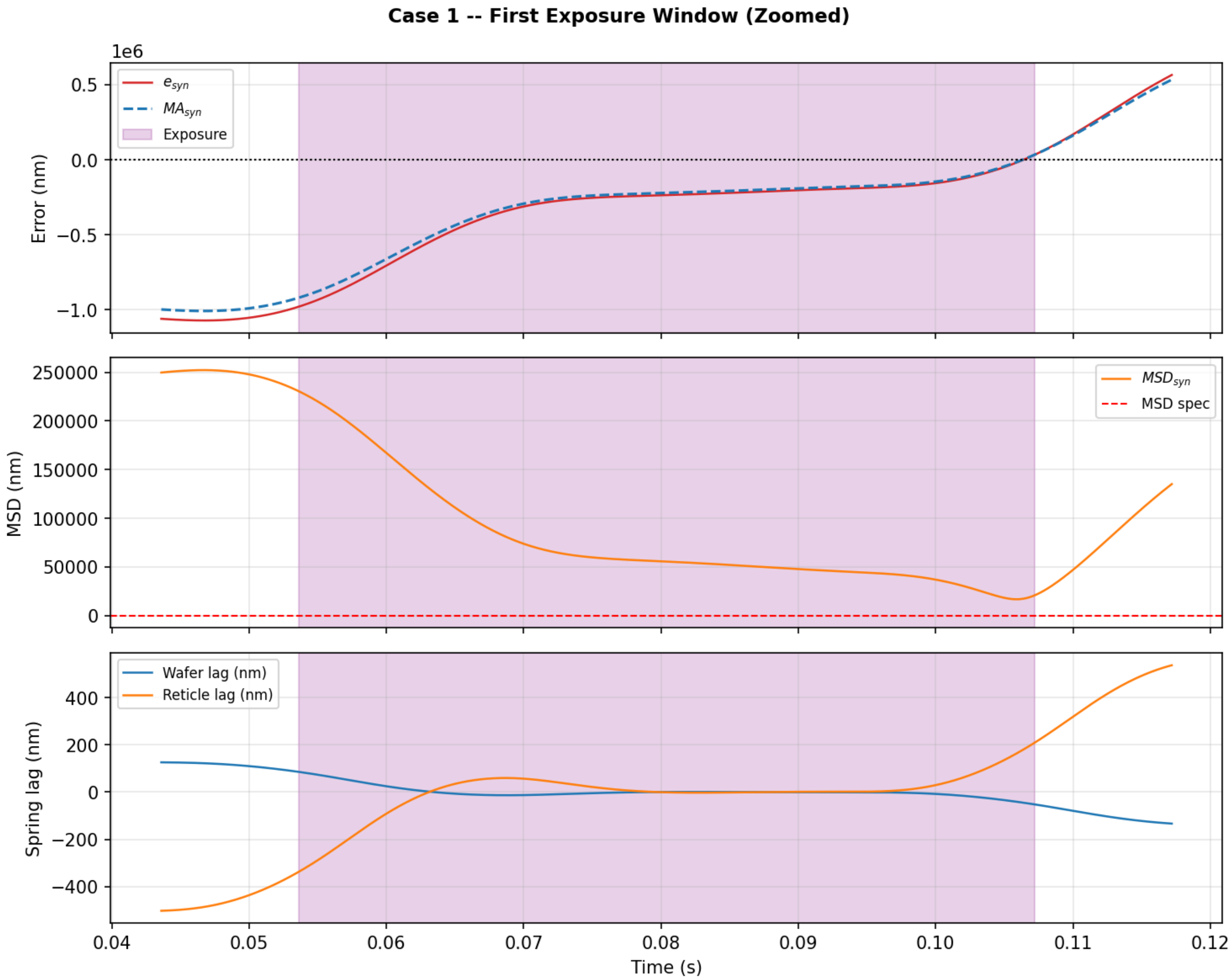


Figure 5.5: Detail of the first exposure window.

the row-normalized confusion matrix.

Table 5.3.1: Per-class evaluation on the test set.

Class	Precision	Recall	F1	Support
none	0.98	1.00	0.99	29,486
Center	0.95	0.90	0.92	859
Donut	0.85	0.84	0.85	111
Edge-Loc	0.87	0.75	0.80	1,038
Edge-Ring	0.94	0.99	0.97	1,936
Loc	0.76	0.63	0.69	718
Near-full	0.96	0.80	0.87	30
Random	0.87	0.76	0.81	173
Scratch	0.00	0.00	0.00	239
accuracy		0.9688		
macro avg	0.80	0.74	0.77	34,590

The confusion matrix exhibits a strong diagonal pattern for eight of the nine classes, with the exception of *Scratch*, which yields zero recall. Given the severe class imbalance of the dataset (239 against e.g. 29,486 *none* samples), the cross-entropy loss is dominated by the majority classes, causing the network to absorb thin-line scratch morphologies into neighboring classes.

5.4 Experiment 3: Trajectory-to-Defect Pipeline

The third experiment evaluates the full traversal of the 300 mm wafer, bridging the physical plant of Experiment 1 with the classifier of Experiment 2. Under long-stroke tracking only, two directions are explored: disturbance injection under fixed trajectory parameters, and a sweep of

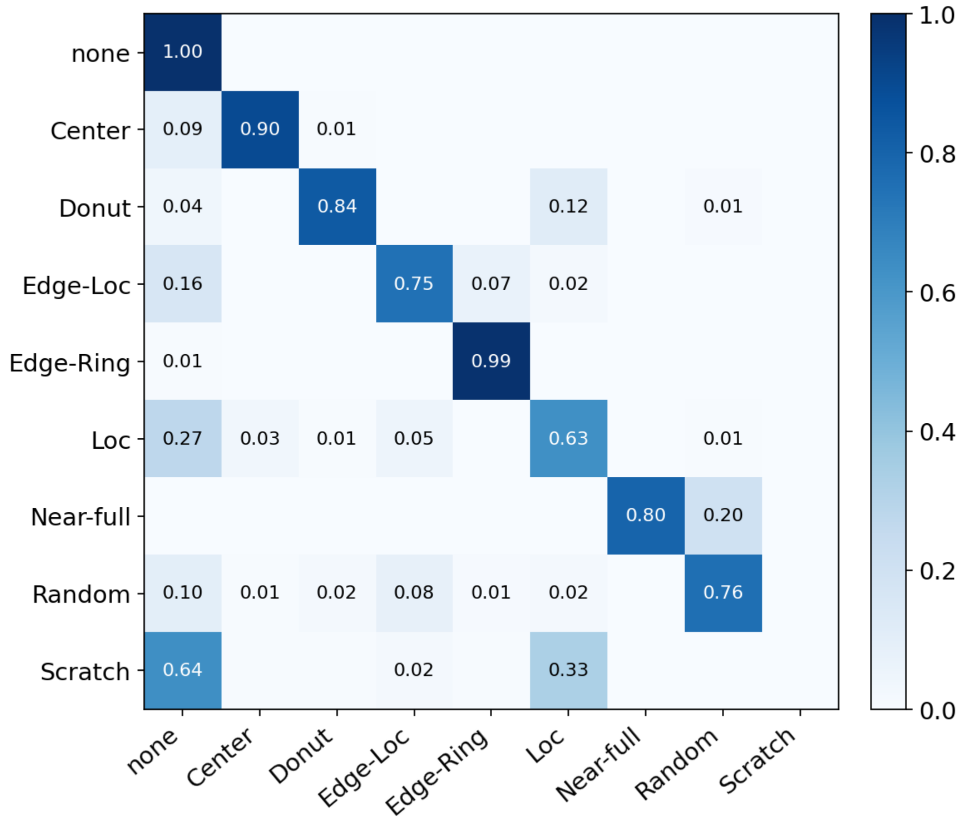


Figure 5.6: Row-normalized confusion matrix.

the kinematic bounds ($v_{w,scan}$, $a_{w,\max}$, $j_{\max}$, $s_{\max}$) under zero disturbance.

With an unactuated short-stroke stage, the exposure-window MSD of the synchronization error lies between 40 and 272 μm depending on the trajectory, nearly four orders of magnitude above the 7 nm specification. Consequently, applying an absolute threshold marks every die as failing, yielding saturated and uninformative wafer maps.

Therefore, for the studies conducted under Case 1 control, the evaluation criterion is the *relative* effect of the disturbance or trajectory parameters, referenced to the unperturbed baseline.

5.4.1 Disturbance signatures

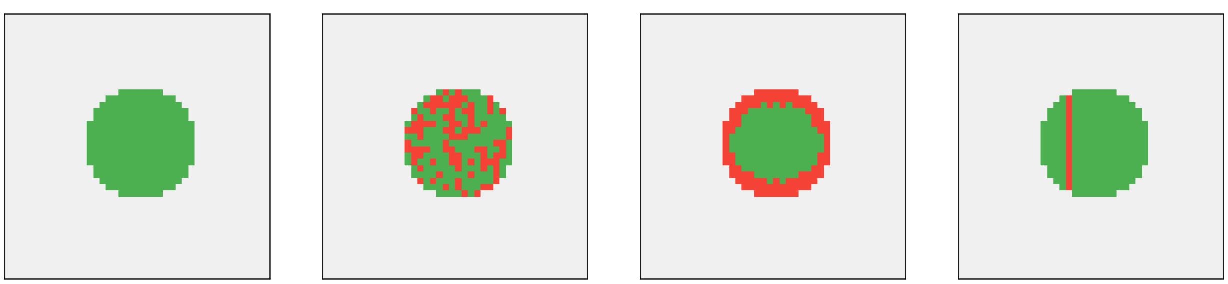
Three disturbance types are injected as forces on the wafer-stage end-effector: broadband force noise on both axes (floor *vibration*), a sustained bias force during scans near the wafer edge (*drift*), and a sustained bias force across a single die column (*scratch*). Synchronization errors are evaluated during the constant-velocity exposure window. A die is marked as failing when its MSD exceeds twice the baseline median or when its mean error shifts by more than 100 nm relative to the unperturbed run.

This campaign uses a fine grid (241 dies of $16 \times 16\,\mathrm{mm}^2$) so that line- and point-like patterns retain their morphology after resizing to the CNN input. Table 5.4.2 and Figure 5.7 present the results.

The spatial component functions as intended since the unperturbed run yields a clean wafer, and each disturbance produces a distinct failure pattern —scattered dies under force noise, an outer ring under edge bias, and a 15-die column under column bias. Nonetheless, the classification performance aligns only partially with the target classes. While the clean wafer is correctly assigned to *none*, the remaining patterns land on morphological neighbors: the ring is classified as *Donut* because the drift band spans wider than a single-die rim; the column is read as

Table 5.4.2: Disturbance-injection campaign.

Disturbance	Failing dies	Spatial pattern	Expected class	CNN prediction
none	0/241	clean wafer	none	none (97.2 %)
vibration	94/241	scattered	Random	Center (40.3 %)
drift	103/241	outer ring	Edge-Ring / Edge-Loc	Donut (99.9 %)
scratch	15/241	one die column	Scratch	Loc (63.1 %)



(a) none (b) vibration (c) drift (d) scratch

Figure 5.7: Simulated wafer maps under the four conditions (yellow: failing, blue: passing).

Loc because classifier recall for *Scratch* is zero; and the scattered pattern (39 % of the wafer) is absorbed into *Center* rather than *Random*.

5.4.2 Kinematic-bound sweep

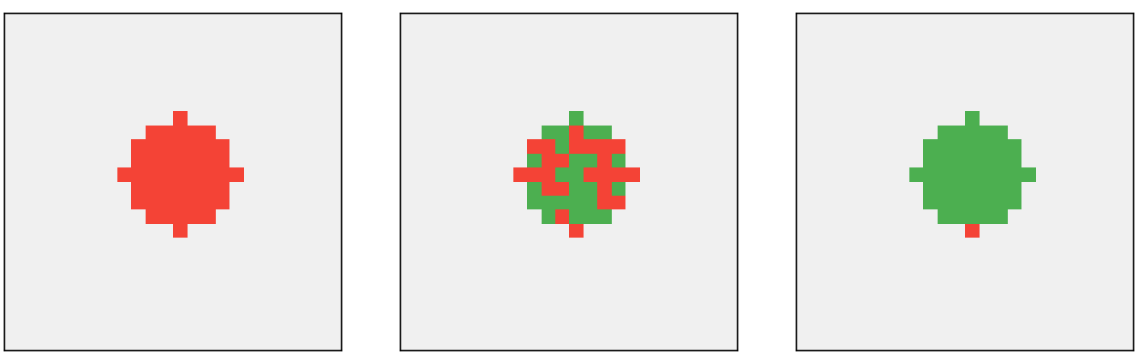
The second campaign sweeps the kinematic bounds around the baseline (0.8 m/s, 20 m/s^2, 1600 m/s^3, 10^5 m/s^4) while holding disturbances at zero on a coarser grid (49 dies of 32×32 mm^2). An aggressive configuration raises all four bounds, a conservative configuration relaxes all four, and four intermediate configurations relax one bound at a time.

Table 5.4.3 summarizes the seven configurations. The median exposure MSD responds monotonically to all four kinematic bounds. Relative to the baseline (101.8 µm MSD, 24 of 49 dies failing), the aggressive configuration causes the entire wafer to fail, driving the median MSD nearly threefold to 272.3 µm. Conversely, the conservative configuration reduces the median MSD by more than half (to 40.0 µm) with only a single failing die. Relaxing any single bound progressively lowers the median error—from 90.2 µm when halving acceleration down to 56.7 µm with a tighter snap—while extending the scan profile to 248 ms compared to 161 ms at baseline. This live snap response is possible because the exact profile generator (Section 4.3) enforces $s_{\max}$ directly rather than deriving it implicitly from ramp duration.

Passing wafer maps (Figure 5.8) yield $P_{\text{none}} = 0.991$ with a residual defect cost of 0.0056. Conversely, the saturated maps under baseline and aggressive bounds concentrate nearly all probability in *Center*—a non-plausible class—causing a naive plausible-class cost to evaluate to zero exactly where wafer quality is worst.

Table 5.4.3: Kinematic-bound sweep.

Config	v [m/s]	a [m/s^2]	j [m/s^3]	s [m/s^4]	T_{scan} [ms]	med. MSD [µm]	Failing dies	CNN top class (prob.)
Baseline	0.8	20	1600	10^5	160.8	101.8	24/49	Center (0.841)
Aggressive	1.2	40	5000	5×10^5	112.6	272.3	49/49	Center (1.000)
Half velocity	0.4	20	1600	10^5	162.9	76.3	1/49	none (0.991)
Half acceleration	0.8	10	1600	10^5	226.3	90.2	4/49	none (0.998)
Quarter jerk	0.8	20	400	10^5	223.5	83.8	4/49	none (0.998)
Tighter snap	0.8	20	1600	10^4	248.1	56.7	1/49	none (0.991)
Conservative	0.2	5	200	5×10^3	333.6	40.0	1/49	none (0.991)



(a) aggressive (b) baseline (c) conservative

Figure 5.8: Simulated wafer maps under the representative setups (yellow: failing, blue: passing).

5.5 Experiment 4: Evaluation of Control Configurations

Experiments prior to this section ran under long-stroke-only control (Case 1), where exposure errors reach the millimeter scale. This section progressively builds up the components of Section 4.4, measuring the contribution of each over the four-die row and making fourth-order trajectories trackable. Table 5.5.4 reports the cruise-window indices of the synchronization error, while Figure 5.9 shows the error traces.

Table 5.5.4: Synchronization error indices under progressive control configurations on the baseline trajectory.

Configuration	max |MA|	max MSD
Case 1: long-stroke PID only	955.0 µm	25.0 µm
Case 2: + lag-compensation FF	99.8 µm	6.54 µm
Case 3a: + short-stroke PI	7.65 µm	0.851 µm
Case 3b: + short-stroke HIGS	7.48 µm	0.834 µm

As established in Section 4.4, under long-stroke-only control (Case 1), the dominant synchronization error is the acceleration-proportional lag of the flexible chain. Adding feedforward compensation (Case 2) cancels this quasi-static lag open-loop (Eq. (4.4.2)), reducing exposure-window MA by nearly an order of magnitude to 99.8 µm. The remaining dynamic transient after each ramp is directly suppressed by closing the loop on the short-stroke stage (Case 3), reducing exposure MSD by a further factor of $\approx 7.8\times$. Overall, full two-stage control yields a combined improvement of $\approx 128\times$ on MA and $\approx 30\times$ on MSD compared to Case 1.

At their respective stability-boundary tunings, the linear PI (Case 3a) and nonlinear HIGS (Case 3b) controllers perform within a few percent of each other, with HIGS marginally ahead on both indices ($\approx 2\,\%$). HIGS expresses its phase advantage as usable gain, sustaining an integrator frequency roughly $42\times$ the PI's equivalent k_i/k_p ratio before destabilizing against the

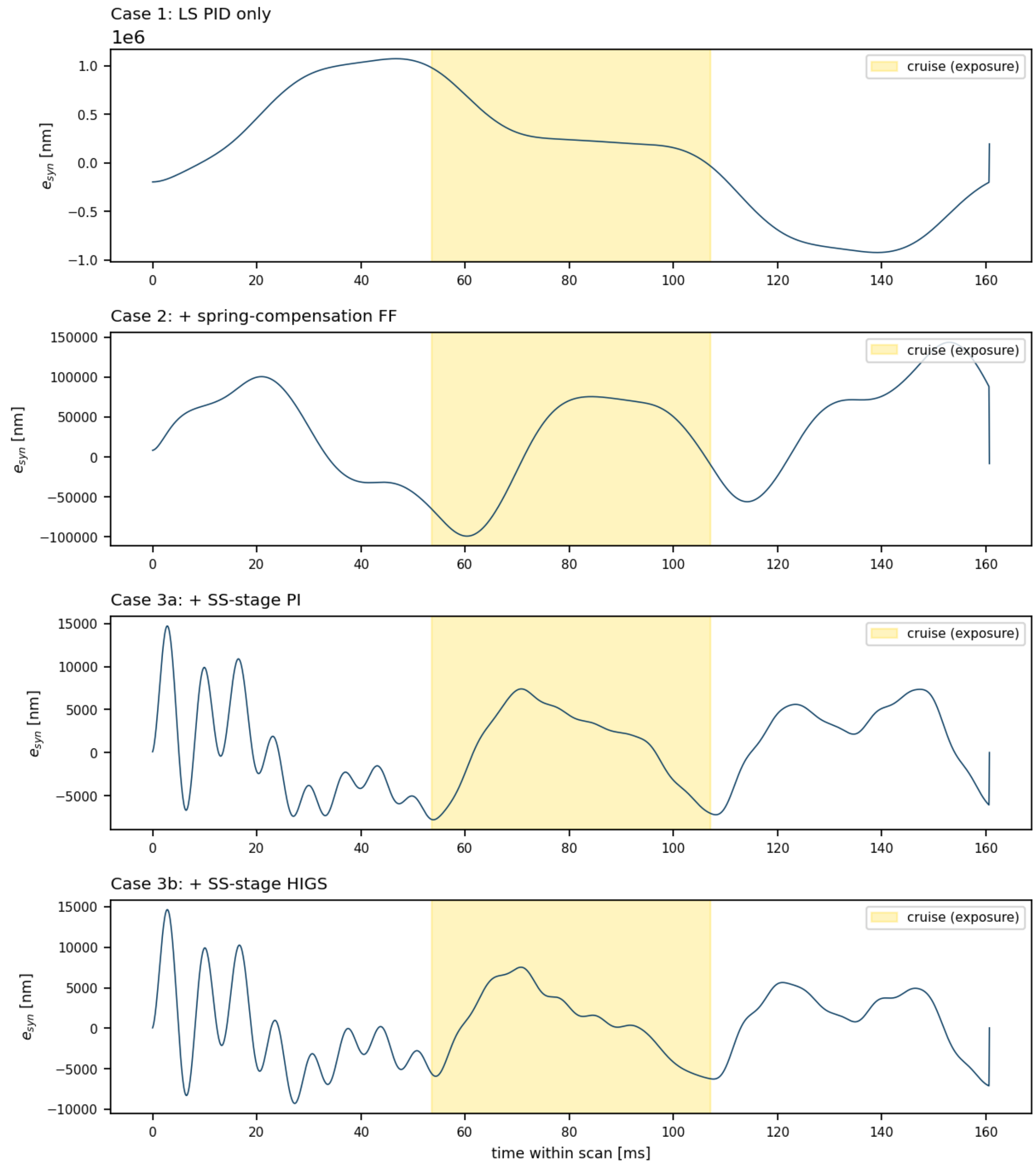


Figure 5.9: Synchronization error during one scan under the four control configurations (shaded band: exposure).

180 Hz passive stage mode ($k_p \approx 16$). However, the primary HIGS advantage in overshoot-constrained settling is not fully isolated by these criteria.

The reference study[2] reports (Figure 14) that the same two-stage architecture reaches an exposure MSD in the tens-of-nanometers range. In contrast, our replication under Case 3b settles roughly two orders of magnitude higher. This disparity stems from the command-following bandwidth of the servo model. Whereas the reference assumes essentially rigid actuator mounts (10^{12} N/m), our plant uses a generic position servo ($K_P = 10^7$ N/m) with an authority capped at its velocity-loop pole $K_P/K_V = 100$ rad/s (≈ 16 Hz), chosen to maintain numerical stability under the implicit integrator. Simply stiffening the mounts does not recover nanometer performance because the authority pole remains fixed; raising K_P merely injects force into the flexible modes, degrading tracking once K_P exceeds a few times 10^7 N/m and stiffening the differential equations beyond integrator limits.

5.5.1 The snap bound versus the exposure specification

For all subsequent experiments, Case 3b serves as the reference controller. Mirroring the spatial setup of Section 5.4.2, this subsection sweeps the snap bound $s_{\max}$ toward increasingly smooth profiles under full two-stage control on a 49-die grid across the full 300 mm wafer. However, every die is now evaluated against the absolute exposure-window specification (Eq. (3.3.3)) to determine whether a fourth-order trajectory can achieve nanometer compliance and what throughput cost it incurs. Table 5.5.5 reports the worst-die indices, failing die count, CNN map classification, and total wafer cycle time.

At baseline ($s = 10^5$), all dies fail and the classifier assigns the map to *Center*. Tightening the snap bound progressively lowers the exposure MSD, but the 7 nm specification is cleared only at $s = 100$ (0 of 49 failing), extending the single-scan duration to 785 ms (48.2 s wafer cycle time). Nonetheless, the moving-average band remains unreached even at the finest tested snap

Table 5.5.5: Snap bound versus the 7 nm exposure-window specification

$s_{\max}$ [m/s^4]	T_{scan}	max |MA|	max MSD	Failing	CNN class	T_{wafer}
10^5 (baseline)	161 ms	7.52 µm	994 nm	49/49	Center	17.7 s
2×10^4	209 ms	2.38 µm	355 nm	49/49	Center	20.0 s
10^4	248 ms	1.27 µm	174 nm	49/49	Center	22.0 s
5×10^3	295 ms	664 nm	63.1 nm	49/49	Center	24.3 s
10^3	441 ms	188 nm	9.5 nm	49/49	Center	31.4 s
100	785 ms	40.5 nm	1.0 nm	0/49	none	48.2 s
50	933 ms	28.7 nm	0.5 nm	0/49	none	55.5 s

($s = 50$), which yields MA = 28.7 nm at a 55.5 s cycle time.

5.6 Experiment 5: The Throughput–Yield Transition

Until this point, sweeping kinematic parameters primarily altered the count of failing dies, as a uniform trajectory stresses every die almost identically. For spatial defect patterns to emerge, dies must experience heterogeneous dynamics. This experiment isolates the primary mechanism responsible for such differences: the settling budget t_{step} between stepping motions. Because exposure begins immediately after each step, the residual transient at scan entry directly contributes to the exposure error. Consequently, dies following atypical stepping moves—above all, row-change dies at the wafer edge that span varying row lengths—enter the scan with larger residuals. As t_{step} shrinks to raise throughput, these edge dies should fail first, yielding a scanner-plausible spatial pattern without injected disturbances.

Shrinking t_{step} from 150 down to 40 ms under a spec-compliant tight-snap trajectory ($s = 100$) results in no failing dies, maintaining an exposure-window MSD of 1.0–1.1 nm throughout. The

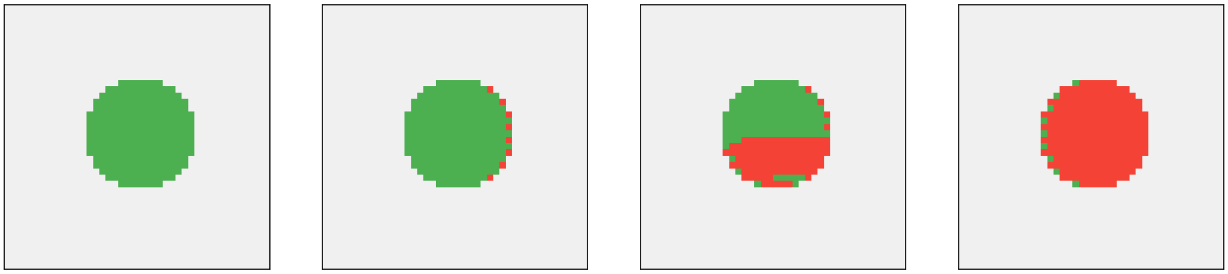
HIGS controller in the short-stroke fine loop settles the post-step transient well within the exposure window even at 40 ms, proving that on this machine, trajectory snap—rather than settling budget—is the binding constraint for absolute specification compliance.

However, as established in Section 5.4.1, referencing the threshold to a trajectory's own settled performance provides a more sensitive lens for identifying *where* failures first emerge. For this sweep under the baseline trajectory ($s = 10^5$), the threshold is set to twice the median exposure MSD of the 200 ms baseline (0.79 µm). In addition to the failing count, two spatial statistics are tracked: the *row-change enrichment* (the failure rate among post-row-change dies relative to the wafer average) and the *scan-order index* (the mean scan position of failing dies, normalized such that 1.0 represents a uniform distribution). Table 5.6.6 reports this onset-referenced sweep across the 241-die grid.

Table 5.6.6: Throughput–yield transition

t_{step}	Failing dies	Row-change enrichment	Scan-order index	CNN class	$\mathcal{J}_{map}$	T_{wafer}
200 ms	0/241	—	—	none (97.2 %)	0.041	82.1 s
190 ms	0/241	—	—	none (97.2 %)	0.041	79.7 s
180–160 ms	8/241	**15.06**	0.94	none (97.4 %)	0.037	77.3–72.5 s
150 ms	103/241	1.17	0.52	Loc (81.9 %)	1.998	70.1 s
120 ms	232/241	0.52	1.00	Center (100 %)	2.000	62.8 s

At $t_{\text{step}} = 180$ ms, the first failures appear as an eight-die cluster with a row-change enrichment of 15.06, meaning every failing die is scanned immediately after a row change. This edge cluster aligns visually with the *Edge-Loc* morphology in the WM-811K dataset[16] and is produced purely by the motion profile. Between 160 and 150 ms, the wafer transitions into a bulk failure of 103 dies, classified as *Loc* rather than *Center*—the classifier's closest reading for a large but partial failure (Section 5.3). Only when t_{step} drops to 120 ms does the map saturate



(a) 200 ms (b) 165 ms (c) 150 ms (d) 120 ms

Figure 5.10: Wafer maps across the throughput–yield transition.

into a *Center*-classified blob (232 failing dies). Across the sweep, wafer cycle time falls from 82.1 to 62.8 s, illustrating the explicit throughput–yield trade: tightening from 190 to 180 ms yields a minor throughput gain before triggering the edge onset, whereas pushing below 160 ms trades modest time savings for severe wafer-wide yield loss.

5.7 Experiment 6: Optimization Scenarios E3 and E4

This experiment bridges the multibody simulation pipeline with the synthetic wafer defect-classification framework. The particle swarm optimization of Section 4.5 is executed twice over the same physical plant, controller configuration (Case 3b), and settling budget ($t_{\mathrm{step}} = 150\,\mathrm{ms}$), with every die evaluated against the absolute exposure-window specification (Eq. (3.3.3)) while varying only the cost function. Both scenarios include a common throughput reward $\gamma\, T_{\mathrm{wafer}}/T_{\mathrm{ref}}$ ($\gamma = 1$, $T_{\mathrm{ref}} = 15\,\mathrm{s}$) so that a faster wafer cycle is favored: scenario E3 employs the conventional wafer-average exposure MSD, whereas scenario E4 incorporates the yield-aware CNN cost (4.5.2). In both scenarios, six particles run for eight iterations (yielding 48 full-wafer simulations, with one particle seeded at the baseline trajectory), where all four kinematic bounds ($v_{\mathrm{max}}, a_{\mathrm{max}}, j_{\mathrm{max}}, s_{\mathrm{max}}$) act as free search dimensions.

Both the third- and fourth-order baselines fail outright, yielding completely failing wafers

Table 5.7.7: Optimization scenarios against the third- and fourth-order baselines at the same operating point.

Scenario	v [m/s]	a [m/s^2]	j [m/s^3]	s [m/s^4]	Failing	CNN class	$\mathcal{J}_{map}$	T_{wafer}
3rd-order baseline	0.80	20.0	1600	—	49/49	Center	2.000	14.5 s
4th-order baseline	0.80	20.0	1600	10^5	49/49	Center	2.000	15.2 s
E3: servo cost	0.67	44.9	5000	170	0/49	none	0.043	37.6 s
E4: CNN cost (seed 0)	0.82	45.0	5000	5×10^5	49/49	Center	2.000	12.4 s
E4: CNN cost (seed 1)	0.77	34.8	5000	4.4×10^5	49/49	Center	2.000	12.8 s

classified as *Center* (Table 5.7.7). Conversely, the search guided by the conventional positioning cost (E3) converges on a trajectory with zero failing dies ($\mathcal{J}_{\text{map}} = 0.043$). The optimizer tightens snap near its lower bound ($s \approx 170\,\text{m/s}^4$) while pushing acceleration and jerk to the upper limits of their search space, yielding a 37.6 s wafer cycle time.

By contrast, the yield-aware CNN-cost search (E4) never leaves the *Center*-saturation plateau with 49 failing dies, across both tested random seeds. Because the corrected cost (4.5.2) assigns every saturated map a uniform ceiling of $\mathcal{J}_{\text{map}} = 2.000$ regardless of failure severity, it provides no optimization gradient to guide the swarm out of saturation. This exposes a fundamental duality in the yield-aware cost, while its flatness on the near side of the specification allows the optimizer to trade acceptable-pattern yield loss for throughput (Sections 5.5.1 and 5.6), that same flatness on the far side becomes a major liability when the swarm starts entirely within the saturated region.

5.8 Experiment 7: Does Trajectory Smoothness Matter Without a Good Controller?

The premise of higher-order trajectory design is that bounding higher-order motion derivatives injects less vibration energy into the mechanical structure. A key research question is whether these trajectory refinements survive under simpler controller architectures: if tracking performance is dominated by low-bandwidth servo lag, does smoothing the reference profile provide any tangible benefit? To investigate this, the experiment crosses profile order across two extremes of control capability: bare long-stroke feedback control (Case 1) and feedforward plus short-stroke HIGS control (Case 3b). All profiles are evaluated on the four-die row under identical kinematic bounds ($v_{\max} = 0.8\,\mathrm{m/s}$, $a_{\max} = 20\,\mathrm{m/s^2}$, $j_{\max} = 1600\,\mathrm{m/s^3}$).

Table 5.8.8: Profile order × controller.

Profile	Controller	T_{scan}	cruise max|MA|	cruise max MSD	full-scan MSD
2nd order	Case 1	132.5 ms	790 µm	53.0 µm	278 µm
3rd order	Case 1	145.0 ms	901 µm	34.5 µm	187 µm
4th order	Case 1	160.8 ms	955 µm	25.0 µm	99.4 µm
4th, tight snap	Case 1	208.6 ms	717 µm	11.9 µm	63.8 µm
2nd order	Case 3b	132.5 ms	53.2 µm	24.0 µm	30.6 µm
3rd order	Case 3b	145.0 ms	15.6 µm	3.27 µm	5.26 µm
4th order	Case 3b	160.8 ms	7.48 µm	0.834 µm	4.59 µm
4th, tight snap	Case 3b	208.6 ms	**2.32** µm	**0.312** µm	4.96 µm

Without active short-stroke compensation (Case 1), the moving-average error remains large across all profile orders due to the unrejected acceleration transient tail, as established in Section 5.2. However, the initial transient magnitude depends directly on reference trajectory

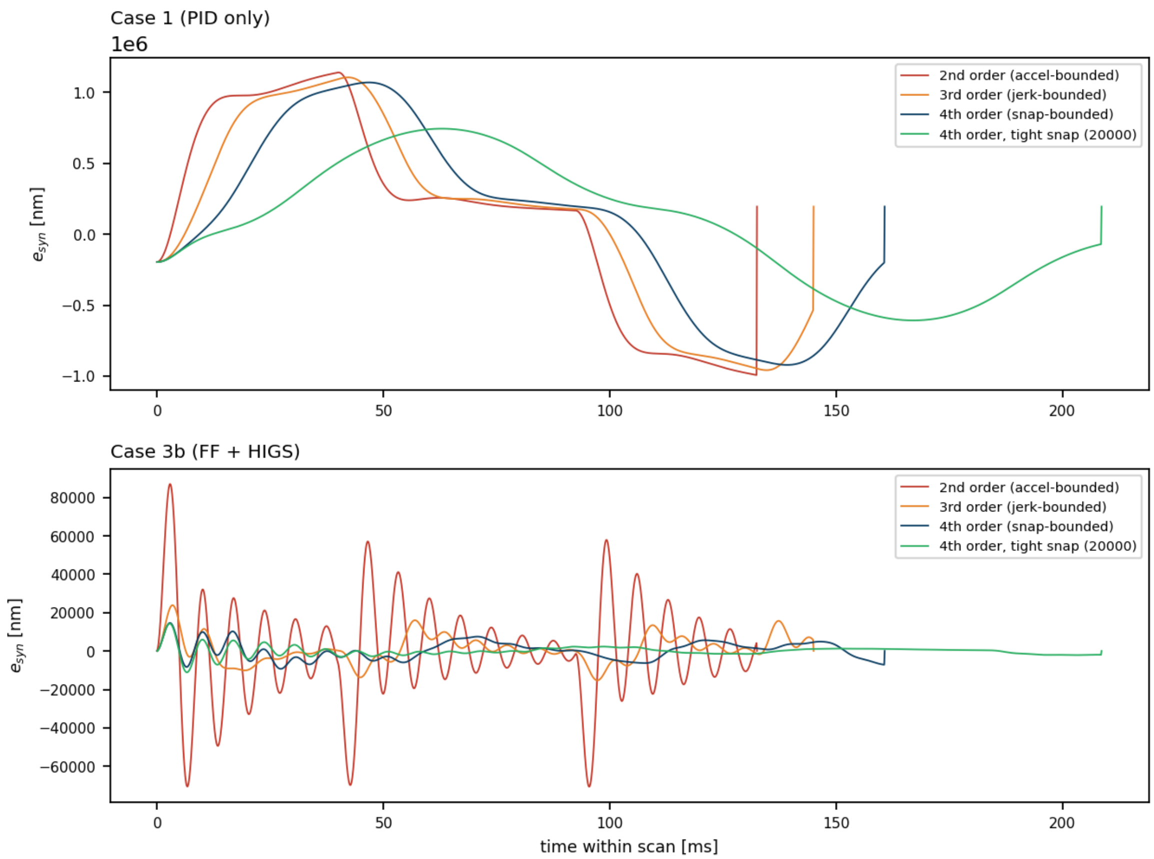


Figure 5.11: Synchronization error during one scan for the four profile orders, under bare Case 1 (top) and Case 3b (bottom).

smoothness. Consequently, cruise MSD improves monotonically with profile order ($4.4\times$ reduction from second-order to tight-snap), and full-scan MSD exhibits a corresponding decrease ($278 \rightarrow 63.8\,\mu\mathrm{m}$). Higher profile order thus achieves a genuine reduction in structural excitation, even though absolute tracking precision remains constrained by uncompensated servo lag.

Under full active control (Case 3b), feedforward cancels quasi-static lag while the HIGS loop suppresses in-band disturbances, allowing reference smoothness to directly reduce out-of-band structural excitation. Consequently, cruise MSD exhibits a $29\times$ error reduction from second- to fourth-order, reaching a $77\times$ reduction under the tight-snap profile. This accuracy gain carries a throughput penalty: scan duration T_{scan} increases by $21\,\%$ for fourth-order and an additional $30\,\%$ for tight-snap. Despite tight-snap achieving the lowest cruise error, its full-scan MSD saturates near the closed-loop entry transient floor, offering no advantage over standard fourth-order despite spending $30\,\%$ more scan time.

Conclusion

This thesis set out to connect two worlds that the wafer-scanner literature treats separately: the servo-mechanical design of reference trajectories, evaluated through settling times and filtered error indices, and the yield engineering of the fab, expressed in the morphology of defect patterns on the wafer. The inverse direction — diagnosing a process cause from an observed wafer fingerprint — is established industrial practice; the forward direction — predicting, from a set of kinematic bounds $(v_{\max}, a_{\max}, j_{\max}, s_{\max})$, the defect pattern they will produce, and optimizing against it — is the gap this work addresses.

The infrastructure for that forward direction was built and validated in simulation. A planar multi-body model of a generic lithography machine, organized as reticle, optics, and wafer chains coupled through Kelvin–Voigt elements, was implemented as a full MuJoCo simulation. Run in its Case 1 configuration (full multi-body machine without fine stages) under collocated PID control with inertia feedforward, it misses the exposure specifications by several orders of magnitude ($\max|\mathrm{MA}| \approx 955\,\mu\mathrm{m}$ against a $2\,\mathrm{nm}$ bound, $\max \mathrm{MSD} \approx 25.0\,\mu\mathrm{m}$ against a $7\,\mathrm{nm}$ bound), with the error dominated by the acceleration-proportional lag that the collocated, long-stroke-only architecture leaves uncompensated downstream of the encoder.

On the classification side, a convolutional network trained on the labelled portion of the industrial WM-811K dataset (172,950 wafer maps) reached 96.88 % hold-out accuracy with a strongly diagonal confusion matrix. Its known weakness — zero recall on the *Scratch* class, a direct consequence of class imbalance — was characterized and its mitigation deferred to the training recipe, where it belongs.

The experiments connecting the two halves produced six concrete findings. First, the exposure-window error responds reproducibly and monotonically to every one of the four kinematic bounds: across the sweep the median exposure MSD scales from 40.0 µm (conservative) through 101.8 µm (baseline) to 272.3 µm (aggressive), and relaxing any single bound (including snap and velocity) lowers the exposure error. Second, the control mechanisms added to the baseline — identified lag feedforward, then a fine loop closed on the measured end-effector error but commanding the short-stroke actuator directly — reduce the exposure error by one and a half to two orders of magnitude (25.0 to 0.834 µm of MSD, 955 to 7.48 µm of MA), with the linear PI and the nonlinear HIGS integrator landing within a few percent of each other at their stability-boundary tunings: with the passive Stage element's 180 Hz resonance sitting near the fine loop's operating range, HIGS's phase advantage provides a marginal edge on cruise tracking at its stability-boundary tuning. Third, and central to the thesis premise: referenced to the trajectory's own settled baseline, the first trajectory-induced failures as the settling budget shrinks concentrate at the row-change dies with fifteen-fold enrichment — an edge-local, scanner-plausible pattern produced by the motion profile alone. Against the absolute 7 nm specification, by contrast, the full controller absorbs the settling transient entirely (no die fails even at a 40 ms budget), so it is the trajectory, not the settle time, that binds — the spatial signature and the spec-compliance are two distinct readings of the same transition. Fourth, trajectory smoothness and controller quality are complements, not substitutes: under a bare PID, profile order reduces transients (4.4× on exposure MSD from second order to tight snap) but cannot reach the exposure window through the dominant transient tail, whereas under full control (Case 3b) the same change of profile order is worth roughly a factor of thirty ($\approx 29\times$, and $\approx 77\times$ to the tight-snap variant) on the exposure MSD at identical kinematic bounds. Pushed further, the snap bound alone walks the exposure error most of the way across the specification: from 994 nm MSD / 7.52 µm MA at the baseline to 1.0 nm MSD (0 of 49 dies failing) at $s = 100$, clearing the MSD half of the bound outright; the tighter moving-average band is not yet crossed at the finest snap tested

($s = 50$: MSD 0.5 nm, MA 28.7 nm), the wafer cycle lengthening from 0.16 to 0.93 s across the tested grid. The MSD specification is thus reachable on this two-stage machine through the *trajectory*, not the servo (raising the actuator-mount stiffness or the loop bandwidth both degrade the tracking); throughput is the price, and the optimum in the snap dimension is interior once that price enters the cost. Fifth, the particle-swarm scenarios closed the loop end to end over the exact snap-bounded profile family and confirmed H3 in the form posed: judged against the absolute 7 nm specification — which both the third- and the fourth-order baselines fail outright, saturating to *Center* — the servo-cost search converges reliably to a spec-compliant, fully clean wafer, confirming again that the snap bound is the dominant lever once it is free to search. The CNN-cost search, with the same standard budget, instead exposed a genuine property of pattern-based costs as optimization landscapes: across two seeds tried, neither escaped the *Center*-saturation plateau, where the cost is pinned at its ceiling regardless of how saturated the map is and offers no gradient to climb out, unlike the servo cost's continuous descent even while every die still fails. Because the compliant snap region occupies a small fraction of the wide search box, 48 evaluations from an unseeded particle swarm are not enough to land inside it by chance. The distinctive lever of the yield-aware cost — spending scanner-plausible yield for throughput — therefore requires either a coarser specification, where the acceptable region is wide enough to find reliably, or a search seeded away from the saturated baseline, a concrete implementation lesson this campaign reinforces more strongly than before. Sixth, exact segment generators for second-, third-, and fourth-order profiles allow every profile order to be evaluated by its true kinematic shape.

The validation criterion for the trajectory-to-defect mapping was met in its spatial and reproducibility halves, and partially in its classification half: unperturbed operation classifies as *none*, the cost shifts reproducibly under parameter sweeps, and structured plausible morphology emerges from the trajectory itself; but the assignment of those patterns to the intended minority classes remains limited by the classifier's scratch recall, sensitivity, and calibration. All observed shortfalls localize in the classifier and cost calibration — bounded refinements — rather than in the forward architecture, which runs end to end: generate, track, map, classify, price, optimize.

The immediate continuation follows directly from the findings. The classifier is the limiting element: retrain it with class rebalancing — above all for *Scratch*, whose recall is zero — and probability calibration, so that motion-induced scratch and edge morphologies are recognized as themselves rather than as their nearest well-populated neighbour. With the pattern classes then reliable, the yield-aware optimization is best exercised at a *coarser node*, where the acceptable-failure window is wide enough for the CNN cost's throughput-for-plausible-yield trade to separate clearly from a servo cost — a trade Experiment 6 showed to be narrow at the tightest 38 nm specification and to widen as the node relaxes. A higher-bandwidth fine-positioning stage, representative of the reference machine's purpose-designed 2 kHz subsystem, remains a worthwhile fidelity extension — not because the nanometer band is otherwise out of reach (the trajectory reaches it, Experiment 3), but because it would lower the throughput price of reaching it and let the settling budget, as well as the snap bound, participate in the trade. A deeper study of the HIGS loop in the overshoot-constrained settling problems where its phase advantage over linear integrators is expected to materialize completes the list.

Bibliography

[1] Yazan M. Al-Rawashdeh, Mohammad Al Janaideh, and Marcel F. Heertjes, *Kinodynamic generation of wafer scanners trajectories used in semiconductor manufacturing*, IEEE Transactions on Automation Science and Engineering **20** (2023), no. 1, 718–732.

[2] Yazan M. Al-Rawashdeh, Mohammad Al Janaideh, and Marcel Heertjes, *On characterization of a generic lithography machine in a multi-directional space*, Mechanism and Machine Theory **170** (2022), 104638.

[3] Hans Butler, *Position control in lithographic equipment [applications of control]*, IEEE Control Systems Magazine **31** (2011), no. 5, 28–47.

[4] Yue Dong, Yu Wan, Shuo Song, and Li Li, *Lithographic stepping trajectory planning for residual vibration suppression: An asymmetric s-curve method*, 2023 International Workshop on Advanced Patterning Solutions (IWAPS), 2023, pp. 1–5.

[5] Jerry H. Ginsberg, *Advanced engineering dynamics*, Cambridge University Press, 2 1984.

[6] {Marcel F.} Heertjes, {Thomas P.} Hoogeveen, and {Sebastiaan J.A.M.} {van den Eijnden}, *Stage motion control: Nonlinear integrators revisited*, Annual Review of Control, Robotics, and Autonomous Systems **8** (2025), no. 1, 267–294 (English), Publisher Copyright: © 2025 by the author(s).

[7] Sung-Ju Jang, Jee-Hyong Lee, Tae-Woo Kim, Jong-Seong Kim, Hyun-Jin Lee, and Jong-Bae Lee, *A wafer map yield model based on deep learning for wafer productivity enhancement*, 2018 29th Annual SEMI Advanced Semiconductor Manufacturing Conference (ASMC), 2018, pp. 29–34.

[8] Jong-Seong Kim, Chang Wook Ahn, Tae-Woo Kim, Hyun-Jin Lee, and Jong-Bae Lee, *A market-oriented wafer map optimization methodology using differential evolution to maximize wafer productivity*, 2017 28th Annual SEMI Advanced Semiconductor Manufacturing Conference (ASMC), 2017, pp. 393–398.

[9] Jong-Seong Kim, Sung-Ju Jang, Tae-Woo Kim, Hyun-Jin Lee, and Jong-Bae Lee, *A productivity-oriented wafer map optimization using yield model based on machine learning*, IEEE Transactions on Semiconductor Manufacturing **32** (2019), no. 1, 39–47.

[10] Auguste Lam, Alexander Ypma, Maxime Gatefait, David Deckers, Arne Koopman, Richard van Haren, and Jan Beltman, *Pattern recognition and data mining techniques to identify*

factors in wafer processing and control determining overlay error, Metrology, Inspection, and Process Control for Microlithography XXIX (Jason P. Cain and Martha I. Sanchez, eds.), vol. 9424, International Society for Optics and Photonics, SPIE, 2015, p. 94241L.

[11] Paul F. Lambrechts, *Trajectory planning and feedforward design for electromechanical motion systems*, Tech. Report DCT 2003-08, Technische Universiteit Eindhoven, 2003.

[12] Paul McLellan, *IEDM 2019: TSMC 5nm*, Cadence Breakfast Bytes blog, 2019.

[13] Robert-H. Munnig Schmidt, *Ultra-precision engineering in lithographic exposure equipment for the semiconductor industry*, Philosophical Transactions of the Royal Society A **370** (2012), 3950–3972.

[14] Luc Tissingh, *Improving wafer stage long stroke commutation using artificial intelligence*, Master's thesis, Eindhoven University of Technology, Eindhoven, Netherlands, 2023, Electrical Engineering.

[15] Emanuel Todorov, Tom Erez, and Yuval Tassa, *Mujoco: A physics engine for model-based control*, 2012 IEEE/RSJ International Conference on Intelligent Robots and Systems, IEEE, 2012, pp. 5026–5033.

[16] Ming-Ju Wu, Jyh-Shing R. Jang, and Jui-Long Chen, *Wafer map failure pattern recognition and similarity ranking for large-scale data sets*, IEEE Transactions on Semiconductor Manufacturing **28** (2015), no. 1, 1–12.

[17] Geoffrey Yeap, S. S. Lin, Y. M. Chen, H. L. Shang, P. W. Wang, H. C. Lin, Y. C. Peng, J. Y. Sheu, M. Wang, X. Chen, B. R. Yang, C. P. Lin, F. C. Yang, Y. K. Leung, D. W. Lin, C. P. Chen, K. F. Yu, D. H. Chen, C. Y. Chang, H. K. Chen, P. Hung, C. S. Hou, Y. K. Cheng, J. Chang, L. Yuan, C. K. Lin, C. C. Chen, Y. C. Yeo, M. H. Tsai, H. T. Lin, C. O. Chui, K. B. Huang, W. Chang, H. J. Lin, K. W. Chen, R. Chen, S. H. Sun, Q. Fu, H. T. Yang, H. T. Chiang, C. C. Yeh, T. L. Lee, C. H. Wang, S. L. Shue, C. W. Wu, R. Lu, W. R. Lin, J. Wu, F. Lai, Y. H. Wu, B. Z. Tien, Y. C. Huang, L. C. Lu, Jun He, Y. Ku, J. Lin, M. Cao, T. S. Chang, and S. M. Jang, *5nm CMOS production technology platform featuring full-fledged EUV, and high mobility channel FinFETs with densest 0.021 μm^2 SRAM cells for mobile SoC and high performance computing applications*, 2019 IEEE International Electron Devices Meeting (IEDM), 2019, pp. 36.7.1–36.7.4.

Appendix A

Full-State Space Model

A.1 State-Space Equations for the Wafer Chain

The wafer chain corresponds to $j = 3$, with $N_3 = 7$ (bodies $i = 2, \ldots, 7$). These map to global state vectors $\mathbf{x}_{2p-1}$ (translational) and $\mathbf{x}_{2p}$ (rotational) for $p = 12, \ldots, 17$. The state-space equations are represented below.

First body of the wafer chain ($i = 2$, $p = 12$):

$$\begin{aligned}\ddot{\mathbf{x}}_{23} &= \frac{1}{\bar{m}_{2_3}}\Big[u^0_{2_3} - {}^0\bar{C}_{2_3}\dot{\mathbf{x}}_{23} - {}^0\bar{K}_{2_3}\mathbf{x}_{23} + {}^0\hat{C}_{3_3}\dot{\mathbf{x}}_{25} + {}^0\hat{K}_{3_3}\mathbf{x}_{25}\Big] \\ &\quad - \frac{1}{m_0}\Big[u^0_1 - {}^0\bar{C}_1\dot{\mathbf{x}}_1 - {}^0\bar{K}_1\mathbf{x}_1 + \sum_{k=1}^{3}\Big({}^0\hat{C}_{2_k}\dot{\mathbf{x}}_{2p_{2_k}-1} + {}^0\hat{K}_{2_k}\mathbf{x}_{2p_{2_k}-1}\Big)\Big] \end{aligned} \tag{A.1.1}$$

$$\begin{aligned}\ddot{\mathbf{x}}_{24} &= \frac{1}{I_{z_{2_3}}}\Big[\tau^0_{2_3} - {}^0\bar{c}_{\theta,2_3}\dot{\mathbf{x}}_{24} - {}^0\bar{k}_{\theta,2_3}\mathbf{x}_{24} + {}^0\hat{c}_{\theta,3_3}\dot{\mathbf{x}}_{26} + {}^0\hat{k}_{\theta,3_3}\mathbf{x}_{26}\Big] \\ &\quad - \frac{1}{I_{z_0}}\Big[\tau^0_1 - {}^0\bar{C}_1\dot{\mathbf{x}}_2 - {}^0\bar{K}_1\mathbf{x}_2 + \sum_{k=1}^{3}\Big({}^0\hat{c}_{\theta,2_k}\dot{\mathbf{x}}_{2p_{2_k}} + {}^0\hat{k}_{\theta,2_k}\mathbf{x}_{2p_{2_k}}\Big)\Big] \end{aligned} \tag{A.1.2}$$

Intermediate body ($i = 3$, $p = 13$):

$$\begin{aligned}\ddot{\mathbf{x}}_{25} &= \frac{1}{\bar{m}_{3_3}}\Big[u^0_{3_3} - {}^0\bar{C}_{3_3}\dot{\mathbf{x}}_{25} - {}^0\bar{K}_{3_3}\mathbf{x}_{25} + {}^0\hat{C}_{(4)_3}\dot{\mathbf{x}}_{27} + {}^0\hat{K}_{(4)_3}\mathbf{x}_{27}\Big] \\ &\quad - \frac{1}{\bar{m}_{(2)_3}}\Big[u^0_{(2)_3} - {}^0\bar{C}_{(2)_3}\dot{\mathbf{x}}_{23} - {}^0\bar{K}_{(2)_3}\mathbf{x}_{23} + {}^0\hat{C}_{3_3}\dot{\mathbf{x}}_{25} + {}^0\hat{K}_{3_3}\mathbf{x}_{25}\Big] \end{aligned} \tag{A.1.3}$$

$$\begin{aligned}\ddot{\mathbf{x}}_{26} &= \frac{1}{I_{z_{3_3}}}\Big[\tau^0_{3_3} - {}^0\bar{c}_{\theta,3_3}\dot{\mathbf{x}}_{26} - {}^0\bar{k}_{\theta,3_3}\mathbf{x}_{26} + {}^0\hat{c}_{\theta,(4)_3}\dot{\mathbf{x}}_{28} + {}^0\hat{k}_{\theta,(4)_3}\mathbf{x}_{28}\Big] \\ &\quad - \frac{1}{I_{z_{(2)_3}}}\Big[\tau^0_{(2)_3} - {}^0\bar{c}_{\theta,(2)_3}\dot{\mathbf{x}}_{24} - {}^0\bar{k}_{\theta,(2)_3}\mathbf{x}_{24} + {}^0\hat{c}_{\theta,3_3}\dot{\mathbf{x}}_{26} + {}^0\hat{k}_{\theta,3_3}\mathbf{x}_{26}\Big] \end{aligned} \tag{A.1.4}$$

Intermediate body $(i = 4,\ p = 14)$**:**

$$\begin{aligned}
\ddot{\mathbf{x}}_{27} &= \frac{1}{\bar{m}_{4_3}}\Big[u^0_{4_3} - {}^0\bar{C}_{4_3}\dot{\mathbf{x}}_{27} - {}^0\bar{K}_{4_3}\mathbf{x}_{27} + {}^0\hat{C}_{(5)_3}\dot{\mathbf{x}}_{29} + {}^0\hat{K}_{(5)_3}\mathbf{x}_{29}\Big] \\
&\quad - \frac{1}{\bar{m}_{(3)_3}}\Big[u^0_{(3)_3} - {}^0\bar{C}_{(3)_3}\dot{\mathbf{x}}_{25} - {}^0\bar{K}_{(3)_3}\mathbf{x}_{25} + {}^0\hat{C}_{4_3}\dot{\mathbf{x}}_{27} + {}^0\hat{K}_{4_3}\mathbf{x}_{27}\Big] \qquad \text{(A.1.5)} \\
\ddot{\mathbf{x}}_{28} &= \frac{1}{I_{z_{4_3}}}\Big[\tau^0_{4_3} - {}^0\bar{c}_{\theta,4_3}\dot{\mathbf{x}}_{28} - {}^0\bar{k}_{\theta,4_3}\mathbf{x}_{28} + {}^0\hat{c}_{\theta,(5)_3}\dot{\mathbf{x}}_{30} + {}^0\hat{k}_{\theta,(5)_3}\mathbf{x}_{30}\Big] \\
&\quad - \frac{1}{I_{z_{(3)_3}}}\Big[\tau^0_{(3)_3} - {}^0\bar{c}_{\theta,(3)_3}\dot{\mathbf{x}}_{26} - {}^0\bar{k}_{\theta,(3)_3}\mathbf{x}_{26} + {}^0\hat{c}_{\theta,4_3}\dot{\mathbf{x}}_{28} + {}^0\hat{k}_{\theta,4_3}\mathbf{x}_{28}\Big] \qquad \text{(A.1.6)}
\end{aligned}$$

Intermediate body $(i = 5,\ p = 15)$**:**

$$\begin{aligned}
\ddot{\mathbf{x}}_{29} &= \frac{1}{\bar{m}_{5_3}}\Big[u^0_{5_3} - {}^0\bar{C}_{5_3}\dot{\mathbf{x}}_{29} - {}^0\bar{K}_{5_3}\mathbf{x}_{29} + {}^0\hat{C}_{(6)_3}\dot{\mathbf{x}}_{31} + {}^0\hat{K}_{(6)_3}\mathbf{x}_{31}\Big] \\
&\quad - \frac{1}{\bar{m}_{(4)_3}}\Big[u^0_{(4)_3} - {}^0\bar{C}_{(4)_3}\dot{\mathbf{x}}_{27} - {}^0\bar{K}_{(4)_3}\mathbf{x}_{27} + {}^0\hat{C}_{5_3}\dot{\mathbf{x}}_{29} + {}^0\hat{K}_{5_3}\mathbf{x}_{29}\Big] \qquad \text{(A.1.7)} \\
\ddot{\mathbf{x}}_{30} &= \frac{1}{I_{z_{5_3}}}\Big[\tau^0_{5_3} - {}^0\bar{c}_{\theta,5_3}\dot{\mathbf{x}}_{30} - {}^0\bar{k}_{\theta,5_3}\mathbf{x}_{30} + {}^0\hat{c}_{\theta,(6)_3}\dot{\mathbf{x}}_{32} + {}^0\hat{k}_{\theta,(6)_3}\mathbf{x}_{32}\Big] \\
&\quad - \frac{1}{I_{z_{(4)_3}}}\Big[\tau^0_{(4)_3} - {}^0\bar{c}_{\theta,(4)_3}\dot{\mathbf{x}}_{28} - {}^0\bar{k}_{\theta,(4)_3}\mathbf{x}_{28} + {}^0\hat{c}_{\theta,5_3}\dot{\mathbf{x}}_{30} + {}^0\hat{k}_{\theta,5_3}\mathbf{x}_{30}\Big] \qquad \text{(A.1.8)}
\end{aligned}$$

Intermediate body $(i = 6,\ p = 16)$**:**

$$\begin{aligned}
\ddot{\mathbf{x}}_{31} &= \frac{1}{\bar{m}_{6_3}}\Big[u^0_{6_3} - {}^0\bar{C}_{6_3}\dot{\mathbf{x}}_{31} - {}^0\bar{K}_{6_3}\mathbf{x}_{31} + {}^0\hat{C}_{(7)_3}\dot{\mathbf{x}}_{33} + {}^0\hat{K}_{(7)_3}\mathbf{x}_{33}\Big] \\
&\quad - \frac{1}{\bar{m}_{(5)_3}}\Big[u^0_{(5)_3} - {}^0\bar{C}_{(5)_3}\dot{\mathbf{x}}_{29} - {}^0\bar{K}_{(5)_3}\mathbf{x}_{29} + {}^0\hat{C}_{6_3}\dot{\mathbf{x}}_{31} + {}^0\hat{K}_{6_3}\mathbf{x}_{31}\Big] \qquad \text{(A.1.9)} \\
\ddot{\mathbf{x}}_{32} &= \frac{1}{I_{z_{6_3}}}\Big[\tau^0_{6_3} - {}^0\bar{c}_{\theta,6_3}\dot{\mathbf{x}}_{32} - {}^0\bar{k}_{\theta,6_3}\mathbf{x}_{32} + {}^0\hat{c}_{\theta,(7)_3}\dot{\mathbf{x}}_{34} + {}^0\hat{k}_{\theta,(7)_3}\mathbf{x}_{34}\Big] \\
&\quad - \frac{1}{I_{z_{(5)_3}}}\Big[\tau^0_{(5)_3} - {}^0\bar{c}_{\theta,(5)_3}\dot{\mathbf{x}}_{30} - {}^0\bar{k}_{\theta,(5)_3}\mathbf{x}_{30} + {}^0\hat{c}_{\theta,6_3}\dot{\mathbf{x}}_{32} + {}^0\hat{k}_{\theta,6_3}\mathbf{x}_{32}\Big] \qquad \text{(A.1.10)}
\end{aligned}$$

End-effector body $(i = 7,\ p = 17)$**:**

$$\begin{aligned}
\ddot{\mathbf{x}}_{33} &= \frac{1}{\bar{m}_{7_3}}\Big[u^0_{7_3} - {}^0\bar{C}_{7_3}\dot{\mathbf{x}}_{33} - {}^0\bar{K}_{7_3}\mathbf{x}_{33}\Big] \\
&\quad - \frac{1}{\bar{m}_{(6)_3}}\Big[u^0_{(6)_3} - {}^0\bar{C}_{(6)_3}\dot{\mathbf{x}}_{31} - {}^0\bar{K}_{(6)_3}\mathbf{x}_{31} + {}^0\hat{C}_{7_3}\dot{\mathbf{x}}_{33} + {}^0\hat{K}_{7_3}\mathbf{x}_{33}\Big] \qquad \text{(A.1.11)} \\
\ddot{\mathbf{x}}_{34} &= \frac{1}{I_{z_{7_3}}}\Big[\tau^0_{7_3} - {}^0\bar{c}_{\theta,7_3}\dot{\mathbf{x}}_{34} - {}^0\bar{k}_{\theta,7_3}\mathbf{x}_{34}\Big] \\
&\quad - \frac{1}{I_{z_{(6)_3}}}\Big[\tau^0_{(6)_3} - {}^0\bar{c}_{\theta,(6)_3}\dot{\mathbf{x}}_{32} - {}^0\bar{k}_{\theta,(6)_3}\mathbf{x}_{32} + {}^0\hat{c}_{\theta,7_3}\dot{\mathbf{x}}_{34} + {}^0\hat{k}_{\theta,7_3}\mathbf{x}_{34}\Big] \qquad \text{(A.1.12)}
\end{aligned}$$